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Pattern Detection in Energy Consumption

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Master’s in Data Science

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Department of Information Sciences and Technologies
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To my family and friends,
for their patience and encouragement.

Acknowledgment

I would like to express my sincere gratitude to my supervisors, PhD Ana Maria de Almeida and PhD Diana Mendes, for their guidance, constructive feedback, and continued support throughout the development of this dissertation.

I also wish to thank the GoiEner energy cooperative for making the smart meter dataset publicly available through Zenodo, enabling this and other data-driven studies of residential energy consumption.

Resumo

A crescente utilização de *smart meters* permite às empresas de energia recolher dados detalhados sobre o consumo de eletricidade ao longo do tempo. Embora estes dados ofereçam informações importantes sobre a forma como a energia é utilizada, apresentam frequentemente anomalias causadas por problemas técnicos, falhas de comunicação ou erros no processamento dos dados. Estas anomalias podem resultar em faturas incorretas, ineficiências operacionais e perda de confiança por parte dos consumidores.

Esta dissertação analisa a forma como padrões e anomalias podem ser identificados em dados de consumo energético provenientes de contadores inteligentes. É realizada uma revisão sistemática da literatura para compreender as abordagens atuais de deteção de anomalias, com especial enfoque em métodos não supervisionados, baseados em previsão e em clustering. Com base nessa revisão, é desenvolvida uma metodologia prática que inclui o pré-processamento dos dados, a análise da sazonalidade, tendência e estrutura residual ao nível da população, e a aplicação de um conjunto de algoritmos de deteção de anomalias sobre séries diárias de consumo agregado por município.

Os métodos são avaliados em três municípios do norte de Espanha pertencentes à cooperativa GoiEner, onde não existem dados de anomalias rotuladas. Os resultados mostram que uma abordagem de ensemble baseada em previsão, combinando N-BEATS com votação maioritária de cinco detetores, produz taxas de anomalia operacionalmente interpretáveis e identifica perturbações de consumo associadas a eventos reais, contribuindo para uma análise de dados mais fiável e uma melhor gestão energética.

Palavras-chave: contadores inteligentes; deteção de anomalias; previsão de séries temporais; consumo de energia; N-BEATS; LSTM; SARIMA; SARIMAX; regressão Ridge; Random Forest; Isolation Forest; Local Outlier Factor; One-Class SVM; K-Means; métodos de ensemble

Abstract

The growing use of smart meters allows energy providers to collect detailed electricity consumption data over time. While this data offers important insights into how energy is used, it often contains anomalies caused by technical problems, communication failures, or data processing errors. These anomalies can result in incorrect bills, operational inefficiencies, and reduced trust from consumers.

This dissertation studies how patterns and anomalies can be identified in smart meter energy consumption data. A systematic review of existing research is carried out to understand current anomaly detection approaches, with a focus on unsupervised, prediction-based, and clustering methods. Based on this review, a practical methodology is developed that covers data preprocessing, population-level analysis of seasonal, trend, and residual structure, and the application of multiple anomaly detection algorithms to daily municipal aggregate consumption series.

The methods are evaluated across three municipalities in northern Spain from the GoiEner cooperative dataset, where no labeled anomaly ground truth is available. The results show that a prediction-based ensemble approach combining N-BEATS forecasting with majority-vote detection across five algorithms produces operationally practical anomaly rates and identifies consumption disruptions that correspond to real-world events, contributing to more reliable energy data analysis and improved operational monitoring.

Keywords: smart meters; anomaly detection; time-series forecasting; energy consumption; N-BEATS; LSTM; SARIMA; SARIMAX; Ridge regression; Random Forest; Isolation Forest; Local Outlier Factor; One-Class SVM; K-Means; ensemble methods

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List of Acronyms

CRISP-DM Cross-Industry Standard Process for Data Mining
IoT Internet of Things
AMI Advanced Metering Infrastructure
SLR Systematic Literature Review
STL Seasonal and Trend decomposition using Loess
PRISMA Preferred Reporting Items for Systematic Reviews and Meta-Analyses
AI Artificial Intelligence
ML Machine Learning
LSTM Long Short-Term Memory
CINN conditional Invertible Neural Networks
CVA conditional Variational Autoencoders
LOF Local Outlier Factor
SVM Support Vector Machine
PCA Principal Component Analysis
GMMs Gaussian Mixture Models
DR Demand Response
CNN Convolutional Neural Network
RNN Recurrent Neural Network
CV Coefficient of Variation
CNAE Clasificación Nacional de Actividades Económicas
MAE Mean Absolute Error
RMSE Root Mean Squared Error
MAPE Mean Absolute Percentage Error
sMAPE Symmetric Mean Absolute Percentage Error
pp percentage point
SARIMA Seasonal Autoregressive Integrated Moving Average
SARIMAX Seasonal Autoregressive Integrated Moving Average with eXogenous inputs
RF Random Forest
N-BEATS Neural Basis Expansion Analysis for Time Series

CHAPTER 1

Introduction

1.1. Context

Energy costs have risen sharply in recent years, and getting billing right depends on having reliable consumption measurements. This dissertation looks at how patterns and anomalies can be identified in smart meter data, and how different algorithms compare at detecting both normal and abnormal consumption behaviour. Smart meters are now widely installed across energy markets, giving retailers and network operators access to large volumes of detailed data recorded at hourly intervals. This level of detail makes it possible to track how households actually use energy, capturing daily habits, demand patterns, and how these shift over time [1]. That level of coverage opens up detection possibilities that single monthly readings cannot support, though individual meter series come with their own challenges [2]. That said, the same data also comes with reliability issues: irregular or faulty readings are common in real deployments and cannot simply be ignored.

These readings frequently show anomalies: abrupt spikes, unexpected drops, missing data, or persistent deviations from a customer's typical history. Such issues often stem from faulty sensors, communication failures, misconfigured hardware, or improper installations rather than actual changes in energy use. When these faults go unnoticed, they cause billing errors that frustrate overcharged customers or create financial losses through undercharging, while also forcing expensive manual inspections [3]. Worse still, undetected anomalies can mask underlying equipment faults that may eventually become safety hazards [2].

Simple rule-based detection is not enough to deal with data this complex. A more useful approach is to first learn what normal consumption looks like, and then flag readings that deviate from it. Smart meter data is well-suited for this: it captures detailed, continuous patterns that reflect real habits and routines. Once a model has learned what typical consumption looks like for a given setting, irregular observations can be identified as genuine deviations rather than just statistical noise. Static thresholds and single baselines tend to miss this nuance [3].

In practice, many systems in the industry still rely on basic thresholds or manual checks that are only triggered after a customer raises a complaint. These approaches do not scale to millions of meters and cannot account for how much energy use naturally varies between people and seasons[3]. Consumption is shaped by weather, time of day, and changing routines, which makes it genuinely hard to separate a real lifestyle change from a faulty reading. There is also no universal definition of what counts as an anomaly, and labelled examples of confirmed faults are rarely available.

There is still no clear consensus on the best way to flag anomalies across different types of energy consumers. Getting this right matters for billing accuracy, operational costs, and consumer trust. Any practical solution needs to handle high-frequency data[4], tolerate noisy measurements, and work without needing labelled fault examples[5].

This dissertation tackles these challenges by studying how to reliably identify abnormal energy consumption patterns using a prediction-based approach. The idea is to model what normal consumption looks like, and then flag days where actual consumption deviates significantly from what was expected. Better detection means fewer billing errors, lower inspection costs, and a more proactive approach to managing the network.

1.2. Motivation

In the energy market, consumption readings determine what customers are charged and form the basis for how retailers manage their networks. Keeping this data accurate and consistent matters for trust between consumers, retailers, and grid operators[3], [6].

Problems arise when readings are wrong. Faulty meters, installation errors, communication failures, and data processing issues can all introduce anomalies that go unnoticed for long periods. When they do, the consequences show up in billing disputes, financial losses, and wasted time on manual investigations that could have been avoided[3].

Smart meters have changed what is possible here. Instead of a single monthly reading, operators now have hourly data for each customer, enough to see daily habits, weekly patterns, and how consumption changes across seasons. Aggregated data misses most of this variation and makes it harder to tell normal from abnormal behaviour [2].

Pattern recognition methods fit naturally into this setting. By learning from historical data what typical consumption looks like, they can flag days where something unexpected happened, without needing a fixed threshold or a labelled example of a fault. Studies applying data-driven methods to energy time series have shown they capture structure that static approaches simply miss [7], [8].

Detecting anomalies automatically and reliably is therefore a practical priority. It directly affects billing accuracy, cuts the cost of manual inspections, and helps operators identify both equipment faults and population-level behavioural disruptions before they affect grid planning or load forecasting. High-resolution smart meter data makes this feasible; the challenge is building methods robust enough to tell apart genuine consumption changes from data errors and equipment faults.

This thesis approaches that challenge at the level of municipal aggregates, not individual meters. Individual smart meter series are too short and too noisy for reliable daily forecasting: on any given day a single user may be away from home, have a broken appliance, or just consume unusually, and those one-off events dominate the signal. Summing consumption across all active meters in a city produces a stable daily series where seasonal and trend structure is clear enough to model properly. Anomalies are then identified as days where that city-level total departs from what the model predicted.

Detecting anomalies at the single-meter level (billing errors on individual accounts, faulty devices) is outside the scope of this framework. That trade-off is discussed in Chapter 5.

1.3. Objectives and Research Questions

Considering the challenges presented in the research,

- RQ1: How are smart meter data readings structured?
- RQ2: In what ways do the seasonality, trend, and residual components behave?
- RQ3: What existing patterns of energy consumption are there?
- RQ4: How do anomaly detection algorithms compare on unlabelled consumption data, and what combination produces operationally interpretable results?

1.4. Research Methodology

The structure of this study follows Cross-Industry Standard Process for Data Mining (CRISP-DM) [9], less as a prescribed workflow and more because the work genuinely unfolded that way. The dataset was messier than expected: around 5,500 users had to be excluded on data-quality grounds before any modelling could begin, and the decision to work with municipal-level aggregates rather than individual meters emerged mid-project, not during the planning stage. A framework that explicitly allows cycling back to earlier phases was a better fit for that kind of discovery-driven process than a strictly linear one would have been.

The most consequential revision during data understanding was the shift away from individual smart meter series. Individual user series are too short and too erratic to forecast reliably at daily resolution: a single user being on holiday or having a broken appliance can dominate the consumption signal for that day. Aggregating across all active meters in a municipality smooths this noise enough to make reliable forecasting tractable. That one decision restructured the analysis more than any algorithmic choice did: it shaped the Seasonal and Trend decomposition using Loess (STL) decomposition work in Section 3.3, justified the municipality selection criteria in Section 3.5, and defined the scope of the anomaly detection framework in Chapter 4.

Deployment is out of scope for this dissertation, but interpretability was a practical consideration throughout. Anomaly rates below 2% and specific flagged dates are numbers a utility analyst could act on; that consideration guided the choice of the $\geq 3/5$ majority vote threshold over looser alternatives that would have produced too many alerts to be useful.

CHAPTER 2

Literature Review

Internet of Things (IoT) sensors, Advanced Metering Infrastructure (AMI), and smart meters have made energy management far more data-intensive over the past decade. Today's meters record consumption at intervals as short as 15 to 60 minutes, producing a continuous stream of readings across residential, commercial, and industrial settings [2]. This wealth of data opens up new ways to detect unusual patterns, whether caused by equipment faults [3], sensor errors, changes in how people behave, operational problems [10], or deliberate actions like electricity theft [11].

Being able to spot these anomalies matters for demand forecasting, load balancing, consumer profiling, and the overall stability of the grid [5]. Yet the problem is genuinely difficult. Consumption patterns vary a great deal depending on weather, building type, and daily routines. And because confirmed fault examples are rarely labelled in real datasets, it is hard to train and validate detection models properly.

2.1. Systematic Literature Review

A systematic literature review Systematic Literature Review (SLR) was conducted to investigate anomaly detection approaches applied to energy consumption time-series data, with particular emphasis on methods operating on smart meter and energy reading datasets. The review aimed to compare different learning paradigms, including unsupervised methods, supervised methods, and clustering-based approaches, as applied to temporal energy data.

2.1.1. Search Strategy

The search strategy followed Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines and was implemented using Scopus, semantic scholar and web of science. A structured search query was employed, combining three conceptual components:

(i) Energy-related data sources ("energy readings," "energy consumption," "smart meters," "energy pattern recognition")
(ii) Temporal data representation ("time series," "time readings")
(iii) Learning paradigms ("unsupervised methods," "supervised methods," "clustering")

(i) Energy-related data sources ("energy readings," "energy consumption," "smart meters," "energy anomaly detection")
(ii) Temporal data representation ("time series," "time readings")

Table 2.1. Pattern Recognition Search Strategy Components

Component	Keywords and Boolean Logic
Data Sources	"energy reading*" OR "energy consumption" OR "smart meter*" OR "energy pattern recognition"
Representation	AND ("time series" OR "time reading*")
Paradigms	AND ("unsupervised" OR "supervised" OR "cluster*")
Number of Results	180
Final String	TITLE-ABS-KEY((C1) AND (C2) AND (C3))

(iii) Learning paradigms ("unsupervised methods," "supervised methods," "clustering")

Table 2.2. Anomaly Detection Search Strategy Components

Component	Keywords and Boolean Logic
Data Sources	"energy reading*" OR "energy consumption" OR "smart meter*" OR "energy anomaly detection"
Representation	AND ("time series" OR "time reading*")
Paradigms	AND ("unsupervised" OR "supervised" OR "cluster*")
Number of Results	165
Final String	TITLE-ABS-KEY((C1) AND (C2) AND (C3))

The combined search strategy was designed to target studies on anomaly detection in energy-related time-series data across supervised, unsupervised, and clustering-based methods.

2.1.2. Screening and Selection

Across both search queries, a combined total of 345 publications were identified. After removing duplicates and screening titles and abstracts, I applied specific inclusion and exclusion criteria. This process narrowed the selection to peer-reviewed studies that focused specifically on energy or smart meter data, utilized explicit time-series analysis, and included an empirical evaluation of their detection methods. After a full-text review, 14 articles were chosen for an in-depth analysis [12]. Figure 2.1 summarises the selection process.

2.1.3. Overview of Included Studies

These selected studies cover several key domains, including building energy management systems, smart meter monitoring, and individual consumer energy profiling. Together, they represent the current state of research regarding anomaly detection techniques for energy time-series data.

Smart meter data availability pushed the field toward hybrid models combining deep learning and clustering; statistical baselines now appear mainly as performance floors. Three problems recur across the reviewed studies: the scarcity of labelled fault examples

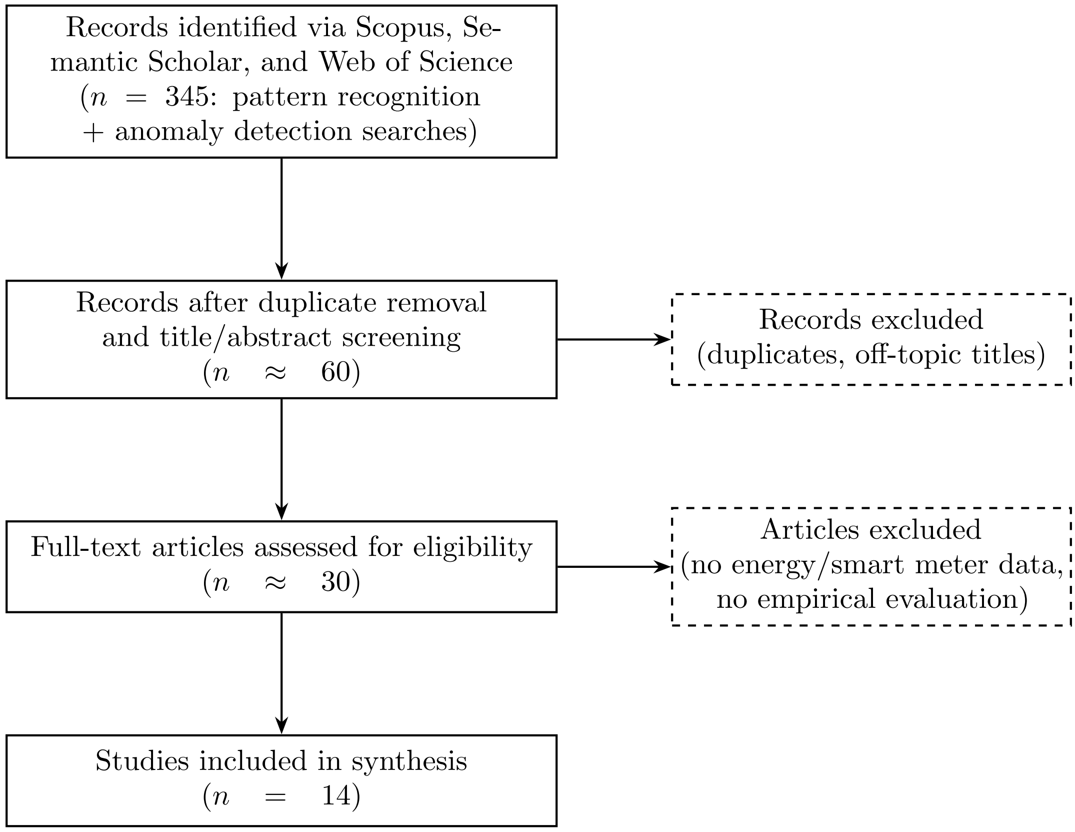

FIGURE 2.1. PRISMA flow diagram for the systematic literature review. Intermediate counts marked ≈ are estimates; exact counts were not separately logged during screening.

[3], the interpretability requirements of operational systems [2], and the heterogeneity of consumption behaviour across households and settings [7].

2.2. Discussion of Related Work

A growing body of research has explored how Artificial Intelligence (AI) and Machine Learning (ML) can be used to detect anomalies in energy consumption data [3]. The widespread adoption of smart meters has made large volumes of high-frequency data available, and there is increasing pressure to use it more effectively to reduce waste and improve grid management [13].

Anomaly detection in this field is typically framed as finding patterns that deviate from what is considered normal [5]. The main difficulty is that normal behaviour varies widely across users and contexts, and confirmed examples of faults or errors are rarely available to train models on [3].

2.2.1. Prediction-Based and Reconstruction-Based Approaches

Prediction-based anomaly detection relies on forecasting models to establish a baseline for "normal" behavior, flagging any significant deviations between the predicted and actual values as anomalies. Transformer-based architectures and deep sequence models have gained traction here because they excel at capturing complex temporal patterns. For instance, Zhang et al. [14] integrated a Transformer forecasting model with K-Means clustering to categorize anomalous residuals, a method that outperformed traditional Long Short-Term Memory (LSTM) baselines in hourly energy datasets. Mao et al. [4] introduced the AF-GS-Random-Forest framework, which uses autoformer predictions and a Random Forest classifier to analyse residuals, reporting improvements in precision, recall, and F1-scores. Outside the scope of the SLR, Shaikh et al. show that N-BEATS applied to a multi-customer smart meter dataset outperforms LSTM, GRU, and temporal convolutional networks on short-term energy consumption forecasting, establishing it as a practical alternative to attention-based architectures when training data is limited [15].

Reconstruction-based methods extend this by operating on latent-space representations rather than direct forecasting. Turowski et al. [10] used conditional Invertible Neural Networks conditional Invertible Neural Networks (CINN) and conditional Variational Autoencoders conditional Variational Autoencoders (CVA) to map raw load curves into a latent space where anomalies become more distinct. Both supervised and unsupervised detection models performed significantly better in this transformed space; representation quality appears to be an upstream bottleneck for detection, not a downstream one.

Prediction and reconstruction models can match that accuracy on large benchmarks, but require large-scale datasets, careful tuning, and substantial compute [2].

2.2.2. Unsupervised Machine Learning and Clustering-Based Algorithms

Unsupervised techniques are the natural choice for anomaly detection in energy data, since real-world datasets rarely include labelled examples of faults. Clustering and density-based algorithms can identify outliers purely from the structure of the data, with no need for pre-labelled examples.

Zangrando et al. [7] performed one of the most thorough comparisons in the field, testing 12 different algorithms across 144 configurations on appliance-level data. Their findings revealed that classical machine learning models, specifically Isolation Forest [16], Local Outlier Factor Local Outlier Factor (LOF) [17], and One-Class Support Vector Machine (SVM), actually performed better than complex deep autoencoders. These traditional models also proved more flexible, generalizing well even when applied to appliances they weren't specifically trained on. Jesmeen et al. found the same on smart home data: LOF ranked among the best methods once feature extraction and Principal Component Analysis (PCA)-based dimensionality reduction were applied [5].

Other clustering frameworks, like Gaussian Mixture Models Gaussian Mixture Models (GMMs), have also proven effective. Liu et al. [11] introduced a multi-cluster feature selection method paired with a GMMs classifier, which significantly improved the detection of abnormal user behavior and electricity theft. Meanwhile, Alonso et al. [18] advanced the field by developing hierarchical clustering algorithms based on quantile autocovariances. Their method allows for the efficient and robust grouping of load profiles on a massive scale.

These properties motivate the selection of Isolation Forest, LOF, and One-Class SVM in the detection pipeline described in Chapter 4.

2.2.3. Statistical, Classical Machine Learning, and Hybrid Methods

A large portion of existing research utilizes statistical and machine-learning techniques that rely on feature engineering and behavioral analysis to spot irregularities. Early methods often focused on regression-based baselines and daily anomaly scoring. For instance, Zhang et al. improved Demand Response Demand Response (DR) predictions by identifying and filtering out abnormal daily load patterns from historical data, ensuring more accurate forecasting [19]. Outside the scope of the SLR, Krishna et al. apply ARIMA to smart meter consumption readings, using forecast prediction intervals as anomaly thresholds and finding that ARIMA captures the non-stationary structure of consumption time series more accurately than simpler autoregressive models [20].

As the field has progressed, researchers have shifted toward classical and deep learning models for analyzing smart meter data. Hernández et al. conducted a comparative study using random forests, decision trees, convolutional neural networks Convolutional Neural Network (CNN), and recurrent neural networks Recurrent Neural Network (RNN) to detect anomalies in daily household routines [1]. Their findings showed that RNN reached an F1-score of 0.8, proving particularly effective at spotting short-term behavioral shifts. Leong et al. showed that Fuzzy C-Means clustering can identify abnormal human activity patterns from energy usage spikes and dips [21].

To handle the sheer volume of data produced by modern grids, hybrid frameworks combining machine learning with big data infrastructure have become more common. Yan et al. developed a system that merges large-scale data preprocessing with deep learning and achieved strong results for real-time operational monitoring [13].

Traditional statistical methods are still used where interpretability or low computational cost matter, but machine learning and deep learning have become the default for energy anomaly detection.

2.2.4. Broader Perspectives and Review Studies

Several review papers offer a high-level look at how this field has evolved. Himeur et al. provide one of the most detailed classifications of AI-driven anomaly detection in energy, pinpointing major hurdles like the lack of labeled data, inconsistent ways of measuring success, privacy issues, and the difficulty of reproducing results across different studies [2]. Gaur et al. point to the same gaps: standardized testing protocols and automated tools for creating ground-truth labels [3].

Other research focuses more on the patterns of energy use itself rather than just the errors. For instance, Rwegasira et al. look at consumption behavior, noting how time-based dynamics, missing data points, and operational changes create a heterogeneous and unstable signal environment that constrains model design [22].

Taken together, these broad reviews highlight systemic problems (scarce labeled data, fragmented methodologies, inconsistent benchmarks) that make comparative studies like this dissertation necessary [2], [3].

2.2.5. Summary

Two findings from the review shaped the design of this study more than anything else. First, classical unsupervised detectors, specifically Isolation Forest, LOF, and One-Class SVM, consistently held their own against deep models on energy data; in Zangrando et al.'s [7] systematic comparison they actually outperformed complex autoencoders on most configurations tested. That result made it defensible to build the detection stage around simpler methods rather than chasing the most sophisticated model available. Second, ensemble voting appeared as a recurring theme across the reviewed studies: no single algorithm dominated across all settings, and cross-method agreement was a more reliable indicator of a genuine anomaly than any individual detector's score [5], [7].

What the literature does not cover well is prediction-based detection at the aggregate level. Most studies operate on individual meters or appliance-level readings, where the noise floor is high and anomalies are defined at fine granularity. Whether a smoothed municipal aggregate series, with its cleaner seasonal structure but a different definition of what counts as anomalous, is easier or harder to monitor is a question none of the reviewed studies address directly. That gap, alongside the absence of any study using the GoiEner cooperative dataset, is part of the motivation for this work [2], [3].

CHAPTER 3

Methodology

This chapter serves a dual purpose. Sections 3.1–3.2 describe the data preparation steps and quality filters applied to the GoiEner dataset (RQ1). Sections 3.3–3.4 present the exploratory analyses of temporal structure and consumption archetypes that answer RQ2 and RQ3, and whose findings directly inform the modelling choices in Chapter 4. Section 3.5 explains the municipality selection procedure that scopes the forecasting and anomaly detection experiments.

3.1. Dataset

3.1.1. Dataset Description

Studying energy consumption patterns requires a dataset that covers a broad range of users over a long enough period to capture seasonal and behavioural variation. For this purpose, the *GoiEner smart meters dataset* [23] was selected. It contains anonymized hourly electricity consumption readings from smart meters managed by the GoiEner energy cooperative in Spain, covering the period from 2014 to 2022.

The eight-year span and hourly resolution make it possible to examine the structure of smart meter readings in detail (RQ1) and to study seasonal, trend, and residual components over time (RQ2). The variety of consumption profiles across users supports the identification of recurring patterns (RQ3), and the presence of real-world data quality issues (missing values and unusual readings) makes the dataset a realistic setting for evaluating anomaly detection methods (RQ4).

3.1.2. Dataset Structure and File Composition

The GoiEner smart meters dataset is setup into multiple files that reflect different stages of data processing and distinct time periods. Each file contains anonymized user hourly electricity consumption time series associated with individual smart meters. The time series are uniformly indexed by timestamp and represent energy consumption measured in kilowatt-hours (kWh), allowing them to be treated as continuous temporal sequences.

The dataset includes both raw and processed versions of the time series. The *raw* data files contain the original smart meter readings as collected from the metering infrastructure. These series may include missing values, irregular gaps, and measurement inconsistencies, reflecting the challenges typically encountered in real-world smart meter deployments. As such, the raw files are particularly useful for assessing data quality, understanding missing data patterns, and motivating preprocessing and anomaly detection strategies.

In addition to the raw data, the dataset provides three imputed subsets corresponding to distinct temporal periods. These files contain time series in which missing values have been filled using a predefined imputation strategy, and an additional indicator is included to distinguish original observations from imputed values. The imputed datasets are segmented into pre-, during-, and post-COVID-19 time periods, reflecting different phases of societal and behavioral conditions.

For analytical convenience, a unified imputed dataset was constructed by concatenating the three imputed subsets into a single continuous time series for each smart meter. This merged file preserves the imputation indicators and temporal ordering of the original subsets, resulting in longer and more complete time series. The unified representation simplifies downstream tasks such as time-series decomposition, forecasting, and prediction-based anomaly detection, which benefit from extended and continuous temporal coverage.

Across all time-series files, each individual record follows a common structure consisting of a timestamp and a corresponding hourly consumption value. In the imputed files, an additional binary flag indicates whether the observation was imputed. This consistent structure enables comparative analysis between raw and processed data, supports time-series decomposition, and facilitates the application of unsupervised anomaly detection algorithms.

Table 3.1. Summary of dataset files and their main characteristics

File	Time Period Covered	Data Status
raw	2014–2022	Not imputed
imp-pre	Before March 2020	Imputed
imp-in	March 2020 – May 2021	Imputed
imp-post	After May 2021	Imputed
merged-imp	2014–2022	Imputed

The merged imputed dataset consolidates all available temporal segments (pre-, during-, and post-periods) for each user into a single time series. As segment availability varies across users, the merged dataset does not enforce complete coverage of all periods but instead preserves all existing observations while explicitly marking segment presence. This representation enables modeling tasks that benefit from extended temporal continuity, such as forecasting and reconstruction-based anomaly detection, while the original segmented files are retained for analyses requiring strict period-specific comparisons.

3.1.3. Metadata Description

In addition to the time-series consumption files, the dataset includes a metadata file that provides descriptive and contextual information for each smart meter time series. This metadata is essential for understanding the composition, duration, and quality of the data, as well as for filtering and grouping time series during analysis. Each row in the metadata file corresponds to a unique anonymized user identifier and summarizes key characteristics of the associated consumption series.

Table 3.2. Description of the metadata file columns

Column	Description
user	Anonymized identifier of the smart meter customer
start_date	Timestamp of the first recorded observation
end_date	Timestamp of the last recorded observation
length_days	Total duration of the time series in days
length_years	Total duration of the time series in years
potential_samples	Expected number of hourly samples for the period
actual_samples	Number of recorded hourly observations
missing_samples_abs	Absolute number of missing samples
missing_samples_pct	Percentage of missing samples
contract_start_date	Start date of the energy supply contract
contract_end_date	End date of the energy supply contract
contracted_tariff	Tariff category associated with the contract
self_consumption_type	Type of self-consumption, if applicable
p1–p6	Contracted power levels for tariff periods
province	Province of the consumer (coarse-grained)
municipality	Municipality of the consumer (coarse-grained)
zip_code	Anonymized postal code
cnae	Economic activity classification code

3.2. Data Understanding

3.2.1. User Filtering and Data Quality

The dataset in this case contains records for 25559 users, however, not all of them are suitable for the purposes of our case study. From the original metadata population, only 19,998 users had successfully generated merged imputed time series and were retained for subsequent modeling and seasonality analyses. Therefore is necessary to assess whether the available data for each user exhibits an adequate composition and quality. For this we use the percentage of missing samples for each time series. This metric allows us to identify users whose data contain substantial gaps and to analyze the overall distribution of missing values across the dataset.

Table 3.3 summarizes the distribution of missing data percentages across all users. The results indicate that an overwhelming majority of users exhibit less than 5% missing samples. Consequently, users with more than 5% missing data can be excluded without distorting the dataset's composition.

After applying this filtering criterion, the resulting dataset comprises 19,709 users. In the following step, we analyze how these users are distributed according to their electricity consumption profiles, with particular emphasis on the number of active tariff periods associated with each user.

Table 3.3. Distribution of missing samples across users

Missing samples range	Count	Percentage (%)
0–5%	19 709	98.55
5–10%	123	0.62
10–15%	46	0.23
15–20%	23	0.12
20–25%	22	0.11
25–30%	12	0.06
30–35%	12	0.06
35–40%	9	0.05
40–45%	8	0.04
45–50%	7	0.04
50–55%	4	0.02
55–60%	5	0.03
60–65%	2	0.01
65–70%	7	0.04
70–75%	2	0.01
75–80%	1	0.01
80–85%	0	0.00
85–90%	0	0.00
90–95%	4	0.02
95–100%	2	0.01

3.2.2. Tariff Period Configuration

An analysis of tariff profile distributions in Figure 3.1 revealed that the majority of users (approximately 89%) are associated with two active tariff periods. Consequently, this group was selected as the primary focus for subsequent analyses. Users with one or six active tariff periods were retained for comparative evaluation, while groups with zero or three active periods were excluded due to insufficient size or incomplete tariff information. Making the total users that are eligible to study being 19,709.

Record duration (years of data per user) varies substantially across profile types and affects how stable any time-series analysis will be. Figure 3.2 shows how this varies across groups.

Users with two active tariff periods have longer records, with a median duration exceeding three years and coverage extending up to seven years. Users with one or six active periods show shorter and more heterogeneous time-series lengths. Based on both group size and temporal coverage, the two-active-period group was selected as the primary population for subsequent analyses, while the remaining groups were retained for comparative evaluation.

The six-active-period group has heavy outliers; duration there is more heterogeneous. The one- and two-period groups are compact with no extreme values. The two-period group is therefore the primary population for all subsequent analyses.

Table 3.4 reports the distribution of contracted tariff period configurations across the filtered user population, expressed in terms of absolute counts and relative percentages.

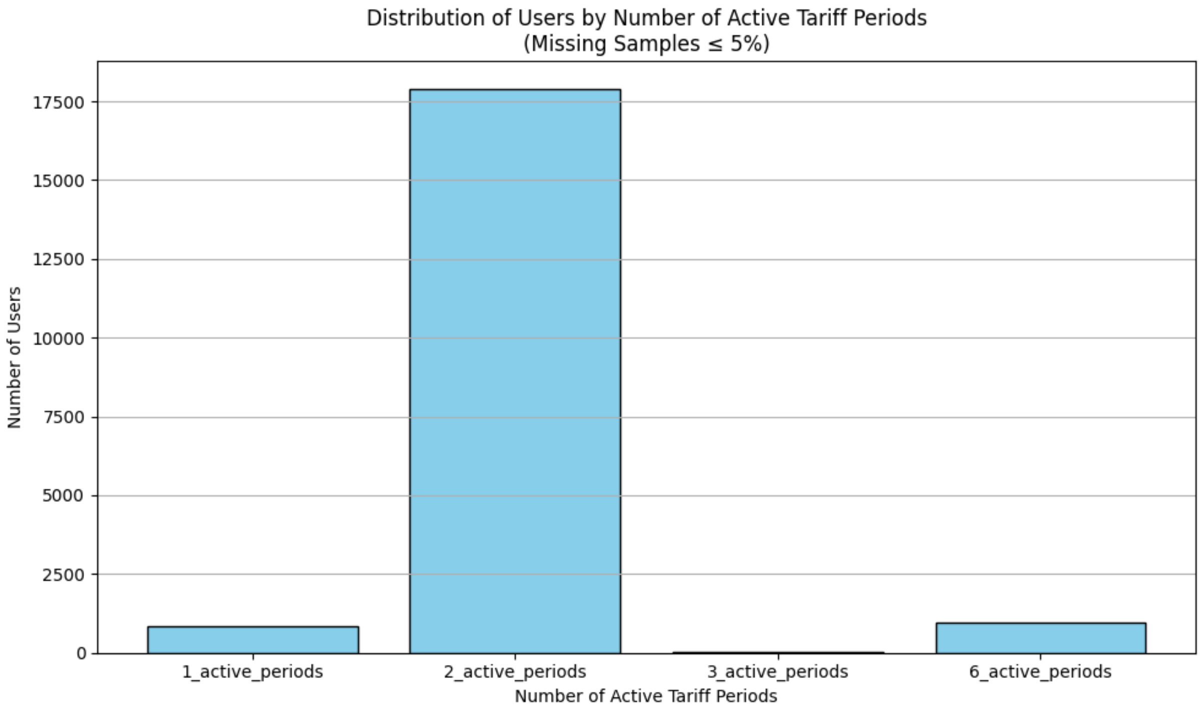

FIGURE 3.1. Distribution of active tariff periods across the dataset.

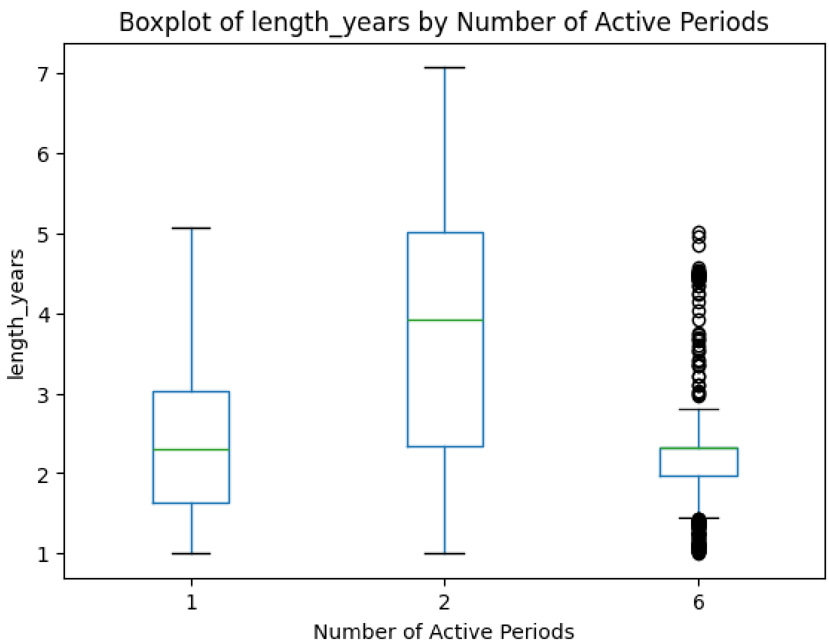

FIGURE 3.2. Distribution of length years by number of active periods.

Each configuration corresponds to a binary representation of the active tariff periods (p1–p6), derived from the contracted power metadata.

The results reveal a highly skewed distribution of tariff configurations. The most common fixture, corresponding to users with active contracted power in periods p1 and p2, accounts for 90.81% of the total population. This configuration reflects the dominant residential tariff structure within the dataset. Single-period configurations, represented

Table 3.4. Distribution of contracted tariff period configurations

Tariff fixture (p1–p6)	Active periods	Users	Percentage (%)
1-1-0-0-0-0	p1, p2	17,870	90.81
1-1-1-1-1-1	p1–p6	939	4.77
1-0-0-0-0-0	p1	855	4.34
1-0-1-0-0-0	p1, p3	15	0.08

by activity in p1 only, comprise 4.34% of users, while users with all six tariff periods active represent 4.77% of the population. Other configurations occur only marginally and account for a negligible proportion of users.

The two-period configuration accounts for over 90% of the filtered population and is the main focus going forward. The remaining configurations are kept for comparison but are too sparsely represented to drive the core findings.

3.2.3. Sector Composition

The distribution of users across Clasificación Nacional de Actividades Económicas (CNAE) sectors is highly uneven, with a strong concentration in a small number of categories. As shown in Table 3.5, the majority of users belong to the Household production sector, which accounts for 16,311 users and dominates the dataset.

All other sectors are much smaller. The second largest group, Public administration, includes only 612 users, followed by Warehousing and logistics (481 users) and Retail trade (346 users). Beyond these categories, most sectors contain relatively few users, with many represented by only a handful of observations.

The dataset is heavily skewed toward residential users. Aggregate results will therefore reflect household behaviour, and sector-specific patterns in smaller groups are likely underrepresented.

The smaller sectors do provide some scope for comparing consumption across economic activities, but sample sizes are too small for firm conclusions.

3.3. Temporal Structure: STL Decomposition

Electricity consumption is inherently periodic. Daily routines, work schedules, and appliance usage impose recurring intraday fluctuations. Understanding this periodicity serves two purposes in this study: it characterises how regular or irregular different user groups are, and it provides the deterministic baseline that is later separated from irregular residuals in the anomaly detection pipeline [5], [7].

3.3.1. Decomposition Methodology

Each user's merged hourly time series is decomposed into trend, seasonal, and remainder components using STL [24]:

$$x_t = T_t + S_t + R_t \tag{3.1}$$

TABLE 3.5. Sample distribution by sector

Rank	Sector	N	%
1	Household production	16,311	80.7
2	Public administration	612	3.0
3	Warehousing and logistics	481	2.4
4	Retail trade	346	1.7
5	Education	211	1.0
6	Food and beverage services	210	1.0
7	Associations	171	0.8
8	Sports and recreation	126	0.6
9	Construction (buildings)	89	0.4
10	Personal services	82	0.4
11	Other professional services	80	0.4
12	Wholesale trade	72	0.4
13	Agriculture, livestock, fishing	71	0.4
14	Water supply	53	0.3
15	Accommodation	45	0.2
16	Real estate	45	0.2
17	Healthcare	44	0.2
18	Specialized construction	37	0.2
19	Vehicle trade and repair	36	0.2
20	Social work	33	0.2
21	Travel agencies	31	0.2
22	Food manufacturing	30	0.1
23	Legal and accounting	30	0.1
24	Other manufacturing	29	0.1
25	Metal products	28	0.1
26	Unknown	22	0.1
27	Office support	20	0.1
28	Libraries, museums	19	0.1
29	Architecture and engineering	18	0.1
30	Financial services	18	0.1
31	Creative arts	16	0.1
32	IT and programming	16	0.1
33	Telecommunications	16	0.1
34	Film, video, TV	15	0.1
35	Printing and reproduction	15	0.1
36	Wood and cork	15	0.1
37	Advertising	13	0.1
38	Machinery	11	0.1
39	Residential care	10	<0.1
40	Energy (electricity, gas)	8	<0.1
41	Others (37 sectors)	144	0.7
	Total	**20,222**	**100.0**

where T_t captures the long-term drift, S_t the periodic pattern, and R_t the irregular residual. Seasonal strength F_s is then computed as:

$$F_s = \max\left(0,\ 1 - \frac{\mathrm{Var}(R_t)}{\mathrm{Var}(S_t + R_t)}\right) \tag{3.2}$$

A value near 1 means the seasonal component explains most of the variation once the trend is removed; a value near 0 indicates little periodic structure. The decomposition is applied at the daily frequency (period = 24 h), independently for the three COVID temporal segments: pre (before March 2020), in (March 2020–May 2021), and post (after May 2021). While weekly periodicity (period = 168 h) is not formally decomposed here, the weekday–weekend cycle is captured explicitly in the forecasting models through calendar features (sin/cos day-of-week encoding and an `is_weekend` binary flag), which are detailed in Section 4.1. Weekly seasonal strength distributions were computed alongside the daily analysis but are not separately plotted; the calendar encoding in the forecasting pipeline is the primary mechanism for capturing the weekday–weekend pattern, and the STL-based weekly metric added no further analytical insight beyond what the daily decomposition already shows.

3.3.2. Seasonality Analysis

3.3.2.1. *Aggregate Daily Seasonality* Figure 3.3 shows the distribution of daily seasonal strength across all users in each period. The in-period median is highest ($F_s = 0.494$), followed by pre (0.462) and post (0.452). Consumption patterns were most regular during the COVID-19 lockdowns and relaxed once restrictions lifted: with mobility restricted and more time spent at home, daily consumption settled into more uniform day-to-day schedules.

The aggregate distributions are positively skewed, with most users clustering around moderate seasonal strength while a smaller group exhibits highly regular behaviour. However, this aggregate view masks important sector-level differences, which are detailed in the sector-stratified analysis below (Table 3.6).

TABLE 3.6. Daily seasonal strength (F_s) by sector and COVID period

Sector	Period	Mean	Median	Std	P25–P75	Skew
Household prod.	Pre	0.476	0.457	0.113	0.406–0.519	+1.42
Household prod.	In	0.501	0.489	0.112	0.433–0.553	+0.97
Household prod.	Post	0.466	0.448	0.112	0.395–0.512	+1.35
Public admin.	Pre	0.863	0.985	0.225	0.782–0.995	−1.58
Public admin.	In	0.831	0.988	0.248	0.699–0.996	−1.25
Public admin.	Post	0.865	0.990	0.228	0.810–0.996	−1.56

3.3.2.2. *Seasonality by CNAE Sector* Each smart meter in the GoiEner dataset carries a CNAE code classifying its user's economic sector. As shown earlier in Table 3.5, the dataset is heavily dominated by the Household production sector ($n = 16{,}311$; 80.7%), with all remaining sectors individually representing less than 3.1% of the population.

To isolate genuine sector-level patterns, the analysis focuses on the two largest groups: Household production and Public administration ($n = 612$). All remaining CNAE categories have sample sizes that are too small for reliable seasonal estimation, a limitation

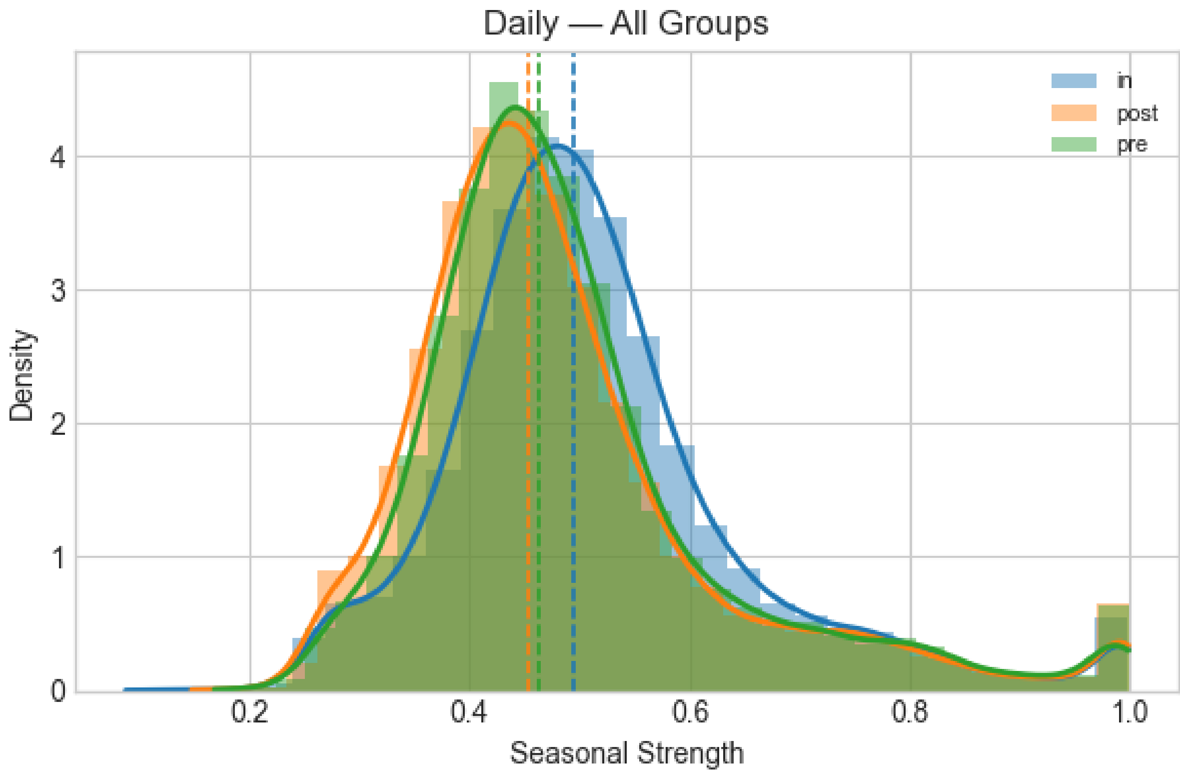

FIGURE 3.3. Distribution of daily seasonal strength (F_s) across all users for the pre-, in-, and post-COVID periods. The in-period distribution shifts visibly to the right, reflecting more structured daily routines during lockdown.

also reported in large-scale smart meter studies that rely on hierarchical grouping to ensure statistical stability [18].

3.3.2.3. *Household Production* The Household production sector exhibits moderate and interpretable daily seasonal strength. Daily F_s values are approximately unimodal and only moderately skewed, as expected for consumers whose routines centre around morning and evening activity peaks [1].

The in-period median rises from 0.457 to 0.489; the post-period (0.448) largely returns to the pre-pandemic level. Lockdown structured household routines more tightly, and that regularity faded as restrictions eased.

3.3.2.4. *Public Administration* The Public administration sector presents a strikingly different daily seasonality profile. As shown in Figure 3.6, the F_s distribution is highly concentrated near 1.0, with median values of approximately 0.985–0.990 across all three periods. This reflects the strong institutional schedules of public buildings, with fixed office hours and administrative calendars that impose a highly regular daily consumption rhythm [1].

Unlike households, the distribution is negatively skewed (skewness ≈ -1.6 in pre and post periods), meaning the bulk of public buildings have near-perfect daily regularity while a small minority of facilities exhibit very low F_s. During the in-period, the mean drops

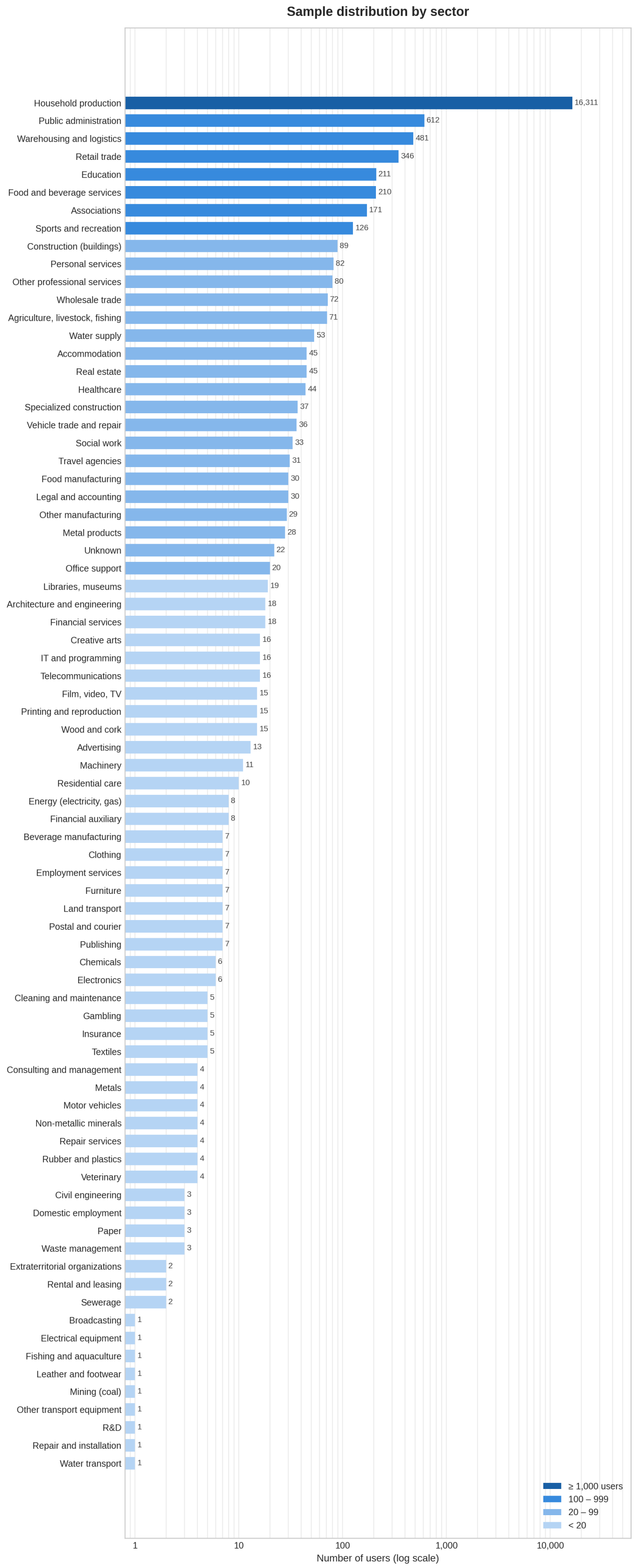

FIGURE 3.4. Distribution of users across CNAE sectors. Household production accounts for the vast majority, while all commercial and industrial categories are substantially smaller.

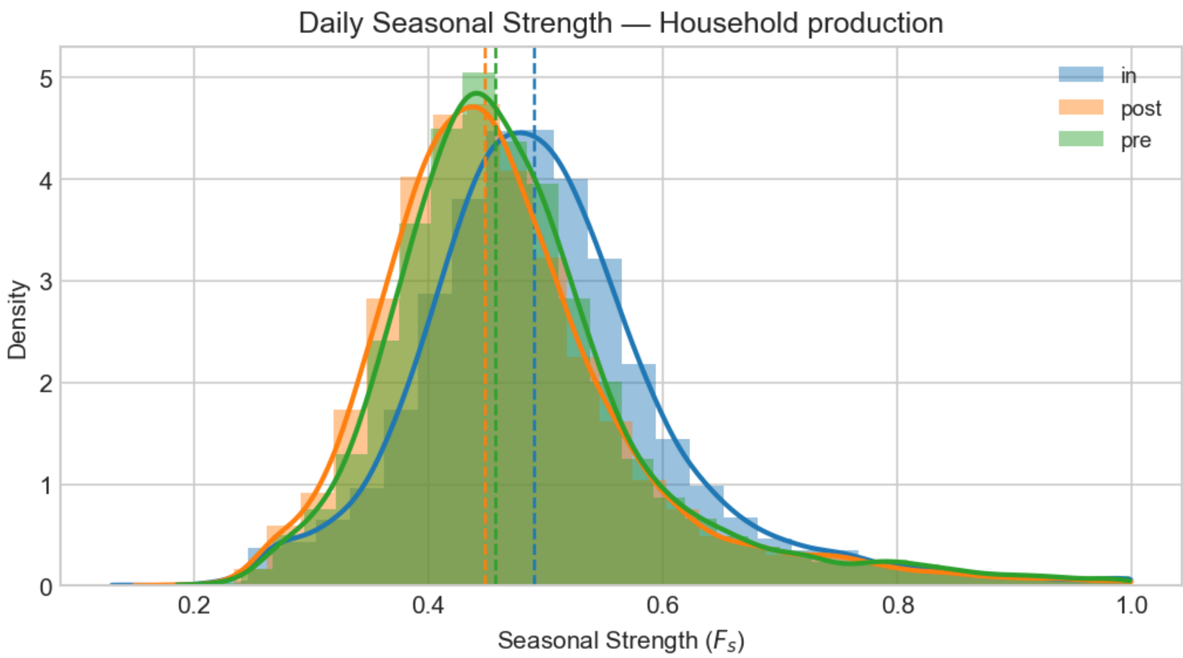

FIGURE 3.5. Daily seasonal strength distributions for the Household production sector, separated by COVID period. The in-period shift toward higher F_s values is clearly visible compared to pre and post.

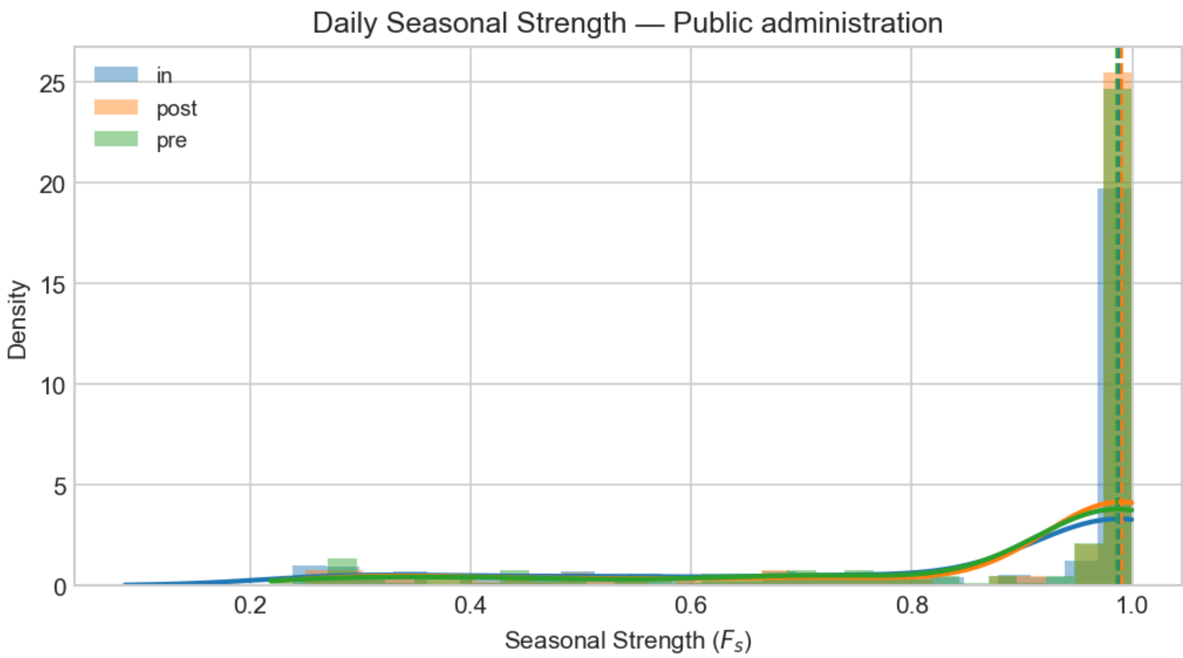

FIGURE 3.6. Daily seasonal strength distributions for the Public Administration sector, separated by COVID period. Most buildings cluster near $F_s = 1.0$, while a minority show very low seasonality.

slightly to 0.831 (from ≈ 0.864) and the standard deviation widens to 0.248. Partial office closures and irregular occupancy during lockdown disrupted daily patterns for a subset of these buildings.

3.3.2.5. *Sector Comparison* The two sectors differ along every dimension of daily seasonality (Table 3.6). Household production shows moderate F_s values (median ≈ 0.45–0.49) with a positive skew, meaning most households are moderately regular and a minority are highly regular. Public administration shows near-perfect regularity for most facilities (median ≈ 0.99) but with a negative skew driven by a minority of buildings with irregular occupancy.

COVID-19 ran in opposite directions across the two sectors: household regularity increased (in-period median 0.489, the highest across periods), but public administration regularity fell (in-period mean 0.831, the lowest), the expected effect of office and public building closures during lockdown. The sector-level divergence also explains the positive skew in the population-wide aggregate: the dominant household group drives the aggregate signal, and the public administration sector's distinct negative-skew contribution is diluted by the large sample-size imbalance [2], [22].

3.3.3. Trend Dynamics

Average trend slopes are near zero in all three periods (Table 3.7), with no aggregate shift up or down. Beneath that stability there is substantial heterogeneity at the user level.

The slope distribution is highly dispersed and right-skewed, especially during the in-period (skewness = 13.46), driven by a small number of users with extreme values. The pre-period distribution is tighter and more balanced; the post-period shows greater spread and a shift toward positive slopes.

The category breakdown confirms it: users with an increasing trend go from 36.4% (pre) to 45.0% (in) and 55.3% (post). The stable group shrinks steadily, and the declining group becomes a smaller minority.

Trend variability remains high across all three periods. Post-period volatility is marginally lower than in the in-period, so the shift toward increasing trends appears gradual rather than sharp.

Table 3.7. Trend Slope and User Behavior Distribution Across Periods

Period	**Mean**	**Median**	**Std**	**Skew**	**Inc.**	**Stable**	**Dec.**
Pre	0.000001	0.000000	0.000014	-5.168	36.4%	36.2%	27.5%
In	0.000004	0.000000	0.000029	13.465	45.0%	19.5%	35.5%
Post	0.000005	0.000002	0.000032	3.615	55.3%	17.8%	26.9%

3.3.4. Residual Structure and Irregular Behavior

The residual component captures whatever the trend and seasonal model cannot explain. For each user, residual variance, Coefficient of Variation (CV), kurtosis, and the proportion of extreme events (observations exceeding three standard deviations) were computed across the three COVID periods.

Figure 3.9 presents the distribution of extreme residual frequency across the population.

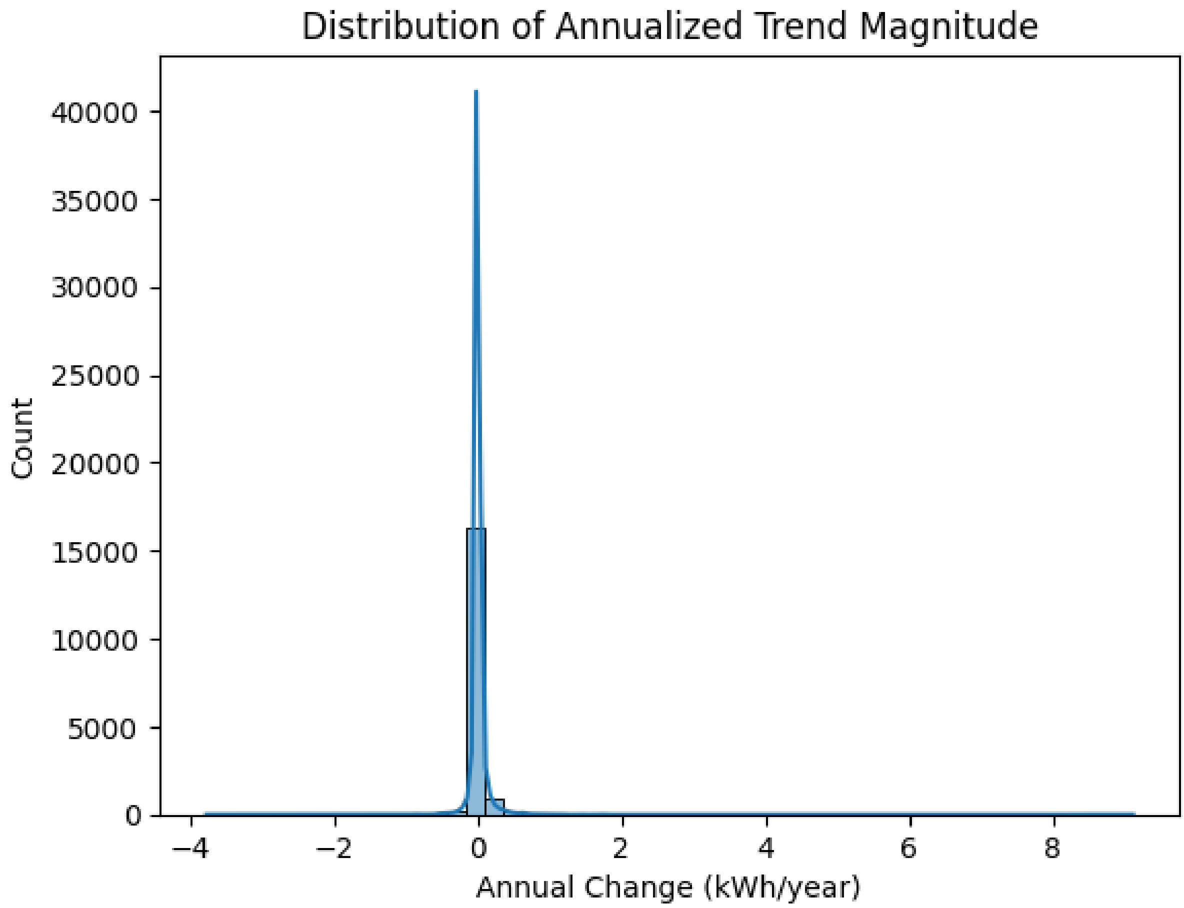

Figure 3.7. Distribution of annualised trend slopes across users, by period. The in-period tail is the heaviest (skewness = 13.46), driven by a small number of users with extreme slope values during lockdown.

The extreme-event rate is almost identical across periods: mean 2.22% (pre), 2.18% (in), 2.18% (post), with medians clustered around 2.26%–2.28%. That stability across very different societal conditions suggests the rate is structural rather than event-driven.

Residual variance does shift, if modestly. The post-period mean (0.0463) is slightly above the in-period (0.0443) and pre-period (0.0437), but standard deviation stays roughly constant. Post-period residuals are somewhat more spread out without being dramatically noisier.

Mean CV rises from 1.037 (pre) to 1.073 (in-COVID) to 1.088 (post), and the median follows the same direction: 0.740, 0.705, 0.751. The upper tail of the CV distribution is heavier in the later periods, pointing to a growing minority of users with highly irregular residuals.

Residual distributions are strongly heavy-tailed in all periods. Median kurtosis runs from 15.0 to 16.2, with upper quartiles exceeding 25 in every group; extreme deviations are far more frequent than Gaussian assumptions would predict. Mean kurtosis peaks in the in-period but all three periods look similar: intermittent spikes are a structural feature of the data, not a COVID-specific artefact.

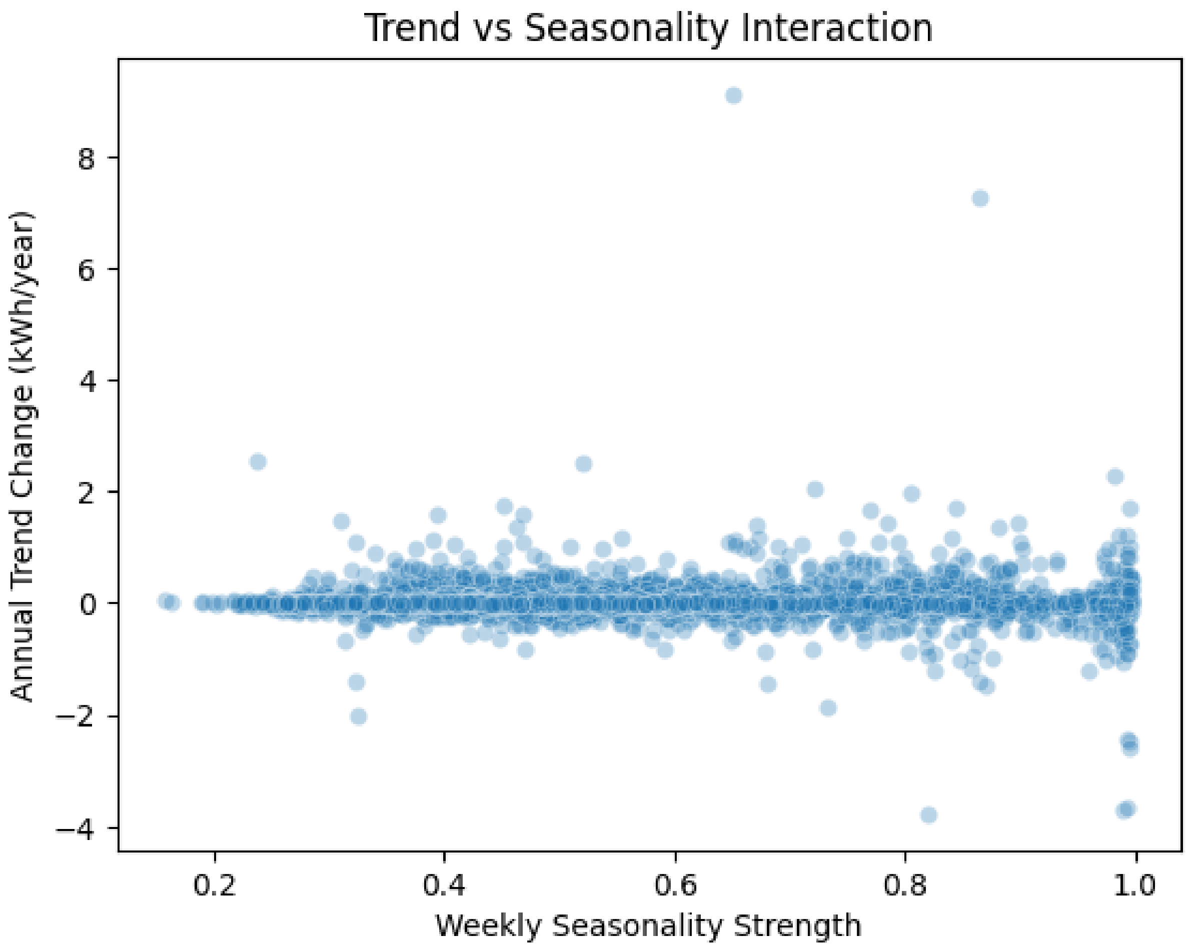

FIGURE 3.8. Trend slope against daily seasonal strength per user and period. No consistent relationship appears between the two: users with strong seasonal regularity are spread across the full range of trend directions, confirming that the two components vary independently.

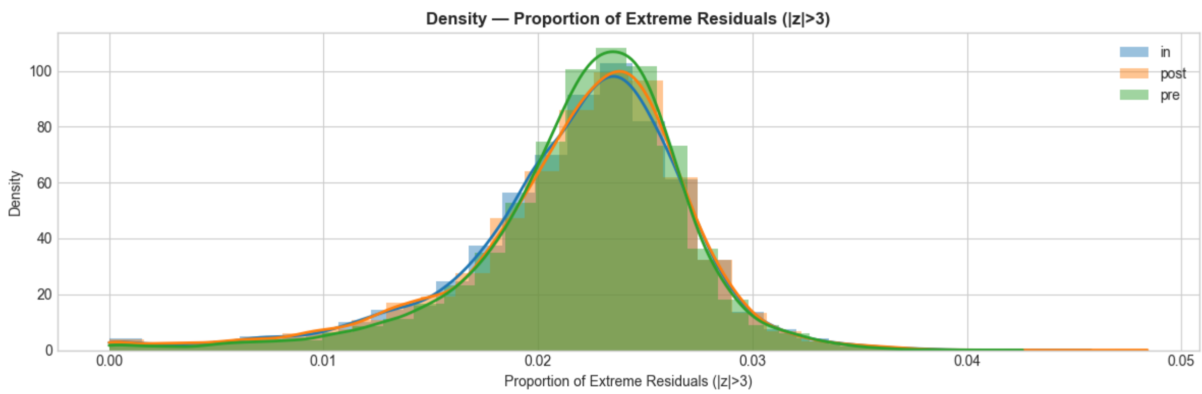

FIGURE 3.9. Distribution of the proportion of extreme residual events across users.

3.4. Consumption Profile Archetypes

The preceding analyses characterise the full GoiEner population: how seasonal, how trended, how noisy. What they do not show is whether users sort into recognisable types

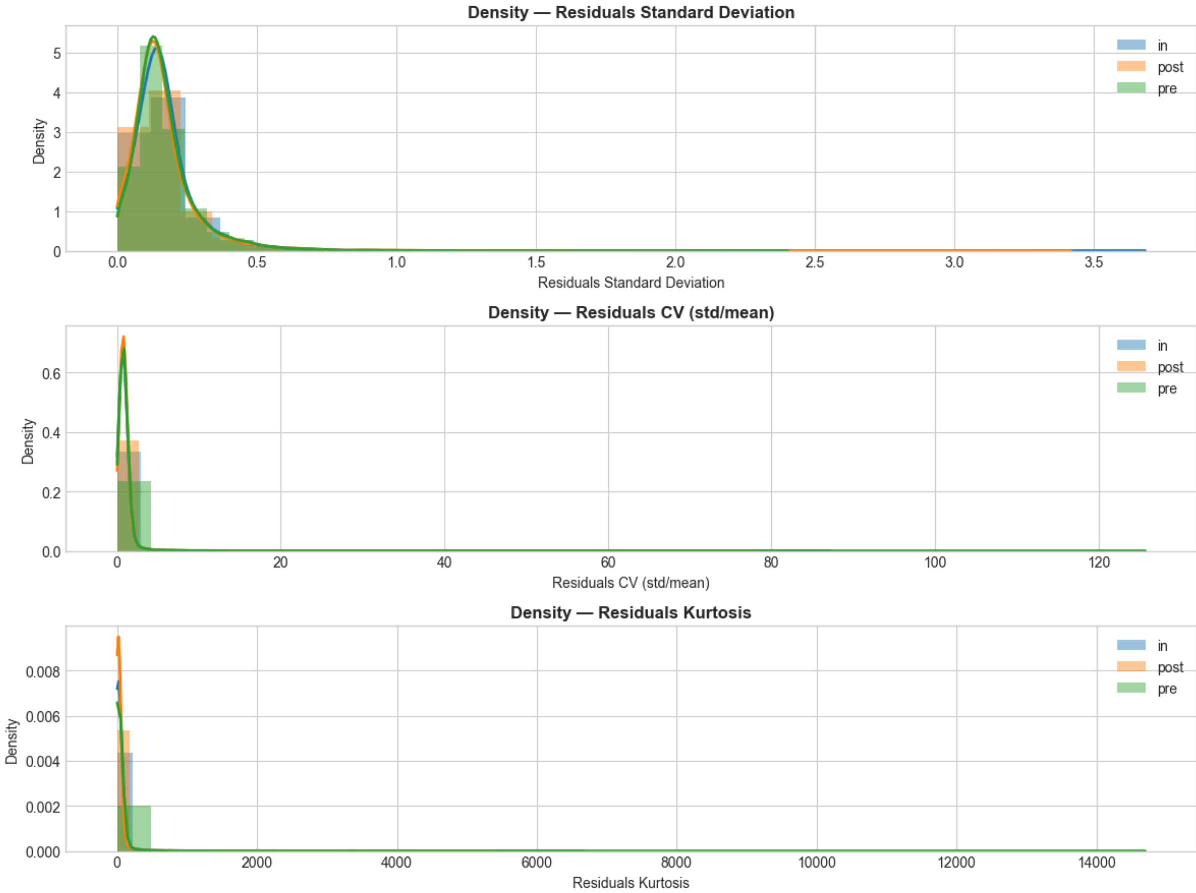

FIGURE 3.10. Distribution of the density of extreme residual events across users.

or whether variation is essentially continuous. To address RQ3 at the user level, k-means clustering [25] was applied to users with complete metrics across all three COVID periods. The GoiEner dataset grew over time as new meters were commissioned: 11,037 users appear in the pre-period, 14,484 in the in-period, and 16,442 in the post-period. Only the 10,015 users (51% of the quality-filtered dataset) present in all three windows can be represented by consistent cross-period averages; the remaining users were excluded from the clustering. Each retained user was represented by four features: average daily seasonal strength (F_s), log-transformed average residual kurtosis, average extreme event ratio, and post-period trend slope. Features were standardised before fitting. Silhouette scores [26] were computed for $k = 2$ through $k = 6$; both $k = 2$ ($s = 0.459$) and $k = 3$ ($s = 0.452$) achieved comparable values before dropping sharply at $k \geq 4$. The three-cluster solution was retained because it separates a high-kurtosis irregular group that $k = 2$ merges into the mainstream, yielding a more actionable segmentation.

Three consumer archetypes emerged, shown in Figure 3.11.

Type A: High Seasonality ($n = 1{,}067$; 10.7%). Strong daily rhythms ($\bar{F}_s = 0.776$) and a predominantly increasing post-period trend (79.5%). The combination of high regularity and rising demand points to premises with fixed daily schedules: office buildings, small commercial outlets, or facilities that returned to, and expanded on, their pre-COVID activity levels.

Type B: Mainstream Residential ($n = 7{,}986$; 79.7%). The large majority of users. Seasonal strength is moderate ($\bar{F}_s = 0.472$), kurtosis is near the population median (mean 19.1), and trends point in no single direction (56% increasing, 28% decreasing, 16% stable). This group reflects the cooperative's dominant household sector: a typical residential consumption profile with no shared direction of change.

Type C: Irregular/Sporadic ($n = 962$; 9.6%). Low seasonal strength ($\bar{F}_s = 0.390$) alongside extremely high kurtosis (mean 330, median 123), yet a *lower* extreme event rate (1.5% vs 2.3–2.4% for Types A and B). These users do not follow a predictable daily rhythm; consumption arrives in concentrated bursts that dominate the kurtosis statistic without inflating the fraction of days with any extreme event. The trend is more often stable or declining (77.7%) than increasing. They likely include vacation properties, storage facilities, or lightly-used commercial premises with intermittent occupancy.

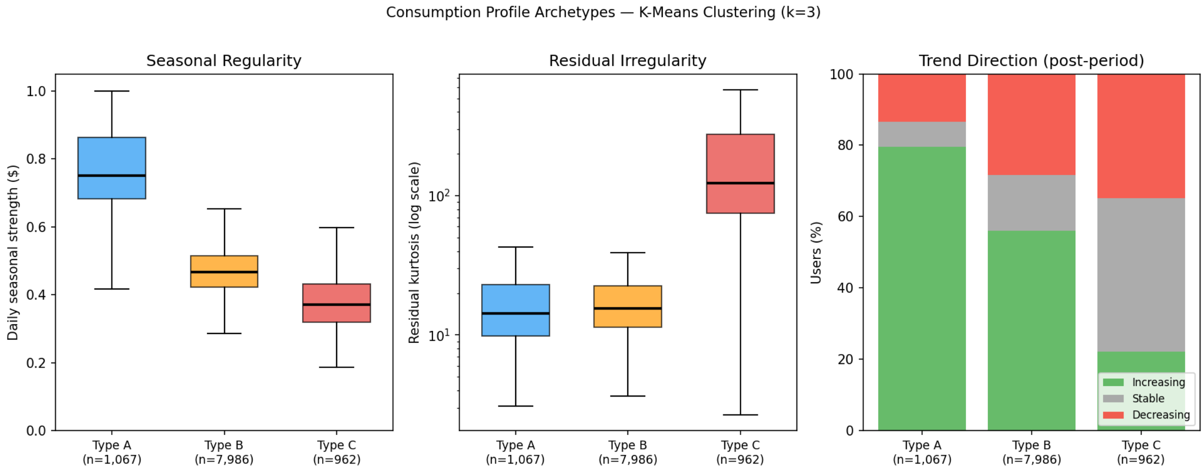

FIGURE 3.11. Consumption profile archetypes from k-means clustering ($k = 3$, $n = 10{,}015$ users). Left: distribution of daily seasonal strength F_s by cluster. Centre: residual kurtosis (log scale). Right: post-period trend direction as a percentage of cluster members. Boxes show the inter-quartile range; outliers are suppressed.

Type B is the GoiEner baseline: moderately seasonal, structurally mixed in trend direction, and typical of a predominantly residential membership. Types A and C are smaller but coherent minorities. Type A grows; Type C is steady or shrinking and highly bursty.

3.5. Municipality Selection

The analyses presented so far operate at the population level, characterising the full GoiEner dataset of approximately 19,700 users. Moving from population statistics to anomaly detection, however, requires a change of granularity. Individual smart meter series are too short and too erratic for the kind of reliable daily forecasting that underpins the prediction-based detection approach: on any given day a single user may be absent, on holiday, or simply consuming unusually, and these one-off events dominate the signal. Aggregating consumption across many users within the same municipality smooths this noise and produces a stable, continuous daily series with well-defined seasonal and trend structure, and therefore a much more tractable forecasting target [18].

3.5.1. Selection Criteria

Not every municipality in the dataset has enough users to make this aggregation meaningful. A municipality whose sample shrinks to ten or twenty users on certain days no longer represents a stable collective signal; it merely reflects the idiosyncratic behaviour of a handful of premises. To guard against this, two inclusion criteria were applied. First, a municipality must have at least 100 active users on any given day, a threshold selected so that random day-to-day fluctuations in active meter counts contribute less than 10% of the variance of the aggregate consumption series, consistent with common practice in large-scale smart meter studies [18]. Second, no more than 10% of its calendar days may fall below that threshold; municipalities with frequent gaps in coverage were excluded to avoid introducing artificial breaks that would confuse the forecasting models.

3.5.2. Selected Municipalities

Three municipalities satisfy both criteria: Vitoria-Gasteiz, Donostia/San Sebastian, and Pamplona/Iruna, all located in the Basque Country and Navarra regions of northern Spain. Beyond meeting the data-quality threshold, these three cities are a deliberate choice: among all qualifying municipalities, they are the one whose mean daily per-user consumption sits furthest above the cooperative-wide average, the one furthest below it, and the one whose mean is most similar to it. Together they provide a varied but coherent test bed for evaluating whether the forecasting and anomaly detection pipeline generalises across cities of different size and consumption character.

Table 3.8 shows the time-based train/validation/test splits applied to each municipality. The boundaries were set so that the validation window sits inside the COVID in-period (where consumption patterns were most disrupted), and the test window falls entirely in the post-period. This design ensures that the models are evaluated under conditions that genuinely differ from their training environment, giving a realistic picture of out-of-sample performance [1].

Table 3.8. Train/validation/test splits by municipality

Municipality	**Training**	**Validation**	**Test**
Vitoria-Gasteiz	2017-01-07 – 2020-10-20 (1,383 days)	2020-10-21 – 2021-08-12 (296 days)	2021-08-13 – 2022-06-05 (297 days)
Donostia/San Sebastian	2017-01-24 – 2020-10-25 (1,371 days)	2020-10-26 – 2021-08-15 (294 days)	2021-08-16 – 2022-06-05 (294 days)
Pamplona/Iruna	2017-05-29 – 2020-12-01 (1,283 days)	2020-12-02 – 2021-09-02 (275 days)	2021-09-03 – 2022-06-05 (276 days)

CHAPTER 4

Results

This chapter reports the empirical results for the three selected municipalities: Vitoria-Gasteiz, Donostia/San Sebastian, and Pamplona/Iruna. The analysis is structured in two sequential stages that reflect the core thesis proposition: that a reliable anomaly detection framework must first establish a high-quality model of normal consumption behaviour before attempting to identify deviations from it.

The first stage (Section 4.1) covers the feature engineering choices, the ten models benchmarked across five families, and their comparative accuracy on the held-out test set. The selection of the best forecasting model is not merely a performance comparison: it determines the quality of the residual signal on which anomaly detection depends. A model that underfits will produce inflated residuals that flag ordinary variation as anomalous; a model that overfits may absorb genuine anomalies into its predictions and suppress the signal entirely.

The second stage (Section 4.2) describes how the best forecaster is used as a baseline to drive five anomaly detection algorithms. The key design insight is that the two stages are tightly coupled: the 2.2% extreme-residual rate observed at the population level in Chapter 3 provides a principled prior for calibrating the contamination parameter of the detectors, and the rolling 30-day z-score replaces the global z-score to keep the anomaly threshold adaptive to seasonal shifts in residual variance. The section compares the sensitivity of each algorithm, examines their cross-method agreement, and identifies the specific days the ensemble flagged across the test window.

All models are evaluated on a common 15% held-out test set covering August 2021 – June 2022, a period entirely within the post-COVID phase. Performance is reported on four standard metrics: Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), Mean Absolute Percentage Error (MAPE), and Symmetric Mean Absolute Percentage Error (sMAPE) [1], [4].

4.1. Forecasting Pipeline

4.1.1. Feature Engineering

Reliable forecasting of aggregate daily consumption requires the model to understand two very different things at once: the rhythms of time (when is it? is today a public holiday? what day of the week is it?) and the momentum of recent consumption (what did demand look like yesterday, last week, a month ago?). Both are captured through 19 engineered features divided into two groups.

The first group consists of nine calendar features that encode temporal context. Rather than treating weekday as a raw integer, which would imply that Sunday and Monday are further apart than Monday and Tuesday, day-of-week and month are encoded as sin/cos pairs so that the calendar wraps around naturally. Three binary flags complete the group: whether the day falls on a weekend, a public holiday, or a bridge day between a holiday and a weekend.

The second group consists of ten history features that capture the momentum and variability of recent consumption: lags at 1, 7, 14, and 30 days, a 7-day and 30-day rolling mean, a 7-day rolling standard deviation, the 7-day rolling ratio, and week-over-week and day-over-day change rates.

4.1.2. Target Variable Selection

Two candidate forecasting targets were evaluated before selecting a final variable:

- `avg_kwh`: the raw daily mean kWh aggregated across all active smart meters in the municipality on a given day.
- `avg_kwh_per_user`: `avg_kwh` divided by the number of active users (n_{users}), yielding a per-capita daily consumption figure.

The distinction matters for two reasons. First, the three municipalities differ substantially in size (Vitoria-Gasteiz is roughly twice as large as Pamplona/Iruna in the GoiEner dataset), so raw aggregates are not directly comparable across cities. Second, the cooperative's user base grew continuously over the 2017–2022 observation window as new smart meters were installed; this growth introduces a spurious upward trend into `avg_kwh` that the models must learn to separate from genuine per-capita consumption dynamics. Dividing by n_{users} removes both sources of confounding, making the three series comparable and stabilising the trend component.

To verify empirically that the normalised target does not degrade forecast accuracy, both targets were run through the same ten-model pipeline under identical hyperparameter settings and the same 70/15/15 train/validation/test split. Table 4.1 reports MAPE and sMAPE for each model and city; MAE and RMSE are excluded because the two targets have different absolute scales and are therefore not directly comparable on those metrics. A negative delta (column ΔMAPE) indicates that `avg_kwh_per_user` yields a lower error, i.e. is the better-fitting target for that model.

Table 4.1. MAPE comparison between `avg_kwh_per_user` and `avg_kwh`

Municipality	`avg_kwh_per_user`		`avg_kwh`		**ΔMAPE (percentage point (pp))**
	MAPE (%)	sMAPE (%)	MAPE (%)	sMAPE (%)	
Vitoria-Gasteiz	38.95	30.48	**5.31**	**5.30**	+33.64
Donostia/San Sebastian	19.50	20.10	**3.38**	**3.40**	+16.12
Pamplona/Iruna	5.78	5.94	**4.42**	**4.44**	+1.36

The empirical comparison reveals that `avg_kwh_per_user` substantially degrades forecasting accuracy on two of the three cities: MAPE increases by 33.6 percentage points on Vitoria-Gasteiz and 16.1 on Donostia/San Sebastian relative to the raw aggregate. This deterioration arises because dividing by the daily active user count creates a very small-valued series (per-user hourly kWh values are typically below 0.5), where small absolute forecast errors produce disproportionately large percentage errors, causing MAPE to become an unreliable optimization signal. Only Pamplona/Iruna, the smallest city with the most stable user base, shows comparable performance across the two targets (ΔMAPE = 1.4 pp).

Based on both the conceptual stability argument and the clear empirical evidence, `avg_kwh` is adopted as the primary forecasting target for all subsequent analysis. All reported metrics, anomaly residuals, and visualisations in this chapter use this series.

4.1.3. Models Evaluated

Ten models spanning five families are benchmarked against each other [1], [4], [14]. The selection follows directly from the literature review in Chapter 2. Prediction-based anomaly detection requires a forecasting model whose residuals serve as the anomaly signal; the review showed that sequence models (LSTM, Transformer variants) and residual block architectures (N-BEATS) consistently outperform statistical baselines on energy time series, while shallow models remain competitive when supplied with strong hand-crafted features [1], [7]. Transformer-based models, although well-represented in the literature, require substantially larger training sets and significantly higher computational resources than were available for three city-level series of approximately 1,300 training days each [2]; they are therefore excluded. Within the neural families, LSTM is included as the dominant sequence architecture in the reviewed studies, N-BEATS as a strong residual block alternative that requires no structural assumptions about the series, and a Seasonal Naive hybrid as a way to encode domain knowledge (annual seasonality) directly into the forecasting structure rather than leaving it for the model to discover. The five baselines provide a performance floor: any model that cannot outperform a 7-day rolling mean or a year-ago copy on a dataset with strong annual structure should not be used for anomaly detection.

- Baselines: Naive (repeat yesterday's value), Seasonal Naive (repeat the value from exactly 365 days prior), Rolling Mean (7-day average), Ridge regression on the engineered features, and Random Forest (300 estimators, max depth 12).
- Statistical: Seasonal Autoregressive Integrated Moving Average (SARIMA) with order $(1, 1, 1) \times (1, 1, 1)_7$, fitted in log-space to handle non-negativity; Seasonal Autoregressive Integrated Moving Average with eXogenous inputs (SARIMAX) with the same differencing order and calendar variables as exogenous regressors.
- Neural networks: LSTM v2, a dual-stream architecture that processes history features through a sequential LSTM [27] (hidden size 48, 2 layers, dropout 0.3) while a static encoder handles calendar context; Neural Basis Expansion Analysis for Time Series (N-BEATS) v2, adapted from the N-BEATS architecture of [28]: unlike the original purely univariate design, this variant conditions three residual

blocks (hidden size 64, dropout 0.2) on the full 19-feature engineering matrix as exogenous context alongside the historical lookback; and Hybrid v2, which uses the Seasonal Naive 365-day forecast as a structural baseline and trains a residual LSTM to correct its systematic errors.

All models are trained on the training split, with hyperparameters and early stopping tuned on the validation split. Final evaluation uses exclusively the held-out test set, which none of the models ever saw during development. Performance is reported on four metrics: MAE, RMSE, MAPE, and sMAPE.

4.1.4. Impact of Feature Engineering

To quantify the contribution of the engineered features, all models were run a second time using only the nine calendar features derived from the date (weekend flag, public holiday flag, bridge-day flag, and the sin/cos encodings of day-of-week, month, and week-of-year). The ten history features (lags, rolling statistics, and change rates) were entirely removed. This creates a controlled ablation: the only difference between the two runs is whether the models have access to consumption history beyond the raw lookback window.

Table 4.2 reports the MAPE for each model under both conditions. The results reveal four distinct behavioural groups.

Traditional models benefit substantially. Ridge regression and Random Forest cannot learn temporal autocorrelation on their own; without explicit lag and rolling features their MAPE climbs to 10–19% across all cities. With the history features the same models drop to 3–8%, a reduction of 7–14 percentage points. This confirms that feature engineering is the primary driver of accuracy for shallow models.

SARIMA benefits substantially from exogenous calendar regressors. The *No FE* column corresponds to plain SARIMA (no exogenous variables); the *With FE* column reports SARIMAX using the nine calendar features as exogenous input. The improvement is large on Donostia (-16 pp) and Pamplona (-21 pp), though Vitoria remains difficult for both (52.99% even with SARIMAX). SARIMA-family models are highly city-dependent.

N-BEATS v2 is feature-agnostic. The differences between the two runs are below half a percentage point on every city ($|\Delta| \leq 0.5$ pp). N-BEATS v2 already learns temporal patterns from its 30-day raw lookback window; explicit lag features add no additional signal.

LSTM v2 and Hybrid v2 are hurt by the extra features. LSTM v2 is 2–3 pp *worse* with feature engineering than without it. The likely cause is redundancy: the sequential LSTM stream already reads the raw lookback window, so feeding it explicit lag and rolling features creates correlated inputs that confuse the gradient signal during training and encourage overfitting. Hybrid v2 shows an even more pronounced deterioration (10–14 pp), because the ResidualLSTM corrector receives the same redundant lag features that are already implicitly captured by the Seasonal Naive component, degrading its ability to model the true residual.

The overall picture is that feature engineering is a net positive for the benchmark as a whole, because the large gains for Ridge, Random Forest (RF), and SARIMAX outweigh the modest losses for the neural sequence models. However, if only neural models are considered, the raw calendar context is sufficient and history features are at best neutral. The feature ablation also provides indirect guidance for future individual-level work: N-BEATS v2's insensitivity to engineered history features suggests its accuracy derives from structural patterns in the raw lookback window rather than hand-crafted lag statistics, a property likely to be more robust to the higher noise levels of individual smart meter series than the feature-dependent models.

TABLE 4.2. MAPE (%) under two feature conditions. Lowest MAPE per column in bold.

	Vitoria-Gasteiz			Donostia/San Seb.			Pamplona/Iruna		
Model	No FE	With FE	Δ	No FE	With FE	Δ	No FE	With FE	Δ
Naive	31.30	31.30	0.00	18.15	22.40	+4.25	16.48	16.48	0.00
Seasonal Naive	12.30	12.30	0.00	10.65	10.68	+0.03	9.25	9.25	0.00
Rolling Mean (7d)	33.31	33.31	0.00	18.07	17.83	−0.23	21.51	21.51	0.00
SARIMA	74.85	52.99	−21.86	19.63	**3.15**	−16.48	27.69	6.23	−21.45
Ridge	15.75	**5.13**	−10.63	17.04	3.53	−13.51	10.93	3.88	−7.05
Random Forest	16.20	7.35	−8.85	19.40	5.78	−13.62	11.72	**3.12**	−8.60
LSTM v2	6.34	9.67	+3.33	**3.45**	6.08	+2.63	4.92	6.82	+1.90
N-BEATS v2	**5.11**	5.31	+0.20	3.85	3.38	−0.47	**4.81**	4.42	−0.39
Hybrid v2	16.39	26.59	+10.20	12.63	27.02	+14.38	9.65	22.62	+12.97

4.1.5. Feature Importance in the Random Forest

Table 4.3 reports the built-in impurity-based importance of the Random Forest trained on the full feature set. Two features absorb 99.7% of split information on average: `pct_rank_global`, the city's daily consumption percentile rank among all municipalities in the dataset, and `zscore_municipality`, its standard-deviation distance from the cross-municipal mean on the same day. All lag and rolling history features, calendar encodings, and day-of-week indicators together contribute less than 0.5%.

Both features are cross-sectional: computing them for day t requires knowing every other municipality's consumption on day t, which is unavailable at forecast time. The Random Forest learns that cities above the daily cross-municipal median have above-average absolute consumption — a dependence on same-day contemporaneous data, not a forecast. The concentration of importance does not mean the model is helpless without these columns: the bootstrap run in Section 4.1.7, which trains Random Forest on calendar and history features only, recovers comparable accuracy because the history features carry largely redundant information. What the importance profile shows is that whenever the cross-sectional shortcuts are available, the model relies on them almost exclusively, so the single-run results in Table 4.4 cannot be read as honest forecasts. This motivates the asterisk exclusion applied there.

For N-BEATS v2 no equivalent importance score is available, but the ablation in Section 4.1.4 already shows the key result: removing all history features shifts N-BEATS MAPE by at most 0.5 pp. The model extracts predictive information directly from the 30-day raw lookback window; the engineered features add nothing.

TABLE 4.3. Random Forest impurity-based feature importance across three cities. The two cross-sectional features absorb essentially all predictive power.

Feature	**Vitoria-G.**	**Donostia**	**Pamplona**
`pct_rank_global`	0.515	0.508	0.515
`zscore_municipality`	0.485	0.483	0.484
All other features (17)	<0.001	0.009	<0.001

4.1.6. Forecasting Results

Table 4.4 reports test-set results. Several patterns stand out. The two simplest baselines (Naive and Rolling Mean) range from 16% to 33% MAPE depending on the city, far above every feature-based and neural model, ruling out recent-history-only approaches for this dataset. Seasonal Naive does substantially better by reusing year-ago values, settling at 9–12% MAPE across the three cities. SARIMA is unreliable: it reaches 74.85% on Vitoria-Gasteiz despite log-space fitting, and even on its best city it trails the neural models. SARIMAX is more competitive and achieves a strong 3.15% on Donostia/San Sebastian, but collapses to 52.99% on Vitoria-Gasteiz, making it too city-dependent to serve as a general solution.

Ridge Features and Random Forest include `pct_rank_global`, the percentile rank of a city's consumption relative to all other municipalities on the *same day*, which requires cross-sectional information that is unavailable at real forecast time. These models are therefore marked with an asterisk and excluded from the best-model comparison. Among the remaining models, N-BEATS v2 achieves the most consistent results across all three cities (MAPE of 5.31% in Vitoria-Gasteiz, 3.38% in Donostia/San Sebastian, and 4.42% in Pamplona/Iruna) and is therefore selected as the base forecaster for the anomaly detection stage. Hybrid v2 is inconsistent (22%–38% across the three cities), and LSTM v2 trails N-BEATS v2 by 2.4–4.4 percentage points. The reliability of these single-run results is assessed in Section 4.1.7.

Table 4.4. Test-set forecasting results by model and municipality. Best values among models free of data leakage in bold.

Municipality	Model	MAE	RMSE	MAPE (%)	sMAPE (%)
Vitoria-Gasteiz	Naive	0.0986	0.1149	31.30	38.69
	Rolling Mean (7d)	0.1045	0.1205	33.31	41.65
	Seasonal Naive (365d)	0.0344	0.0450	12.30	11.79
	Ridge Features*	0.0150	0.0210	5.13	5.13
	Random Forest*	0.0228	0.0264	7.35	7.68
	SARIMA	0.2214	0.2392	74.85	132.38
	SARIMAX	0.1557	0.1691	52.99	77.55
	LSTM v2	0.0296	0.0392	9.67	9.98
	N-BEATS v2	**0.0154**	**0.0207**	**5.31**	**5.30**
	Hybrid v2	0.0640	0.0778	24.30	20.50
Donostia/San Sebastian	Naive	0.0912	0.1090	22.40	26.21
	Rolling Mean (7d)	0.0739	0.0938	17.83	20.47
	Seasonal Naive (365d)	0.0400	0.0523	10.68	10.32
	Ridge Features*	0.0139	0.0193	3.53	3.62
	Random Forest*	0.0228	0.0267	5.78	6.00
	SARIMA	0.0541	0.0732	12.98	14.38
	SARIMAX	**0.0109**	**0.0150**	**3.15**	**3.06**
	LSTM v2	0.0244	0.0328	6.08	6.26
	N-BEATS v2	0.0132	0.0183	3.38	3.40
	Hybrid v2	0.1357	0.1557	37.77	30.38
Pamplona/Iruna	Naive	0.0637	0.0820	16.48	18.74
	Rolling Mean (7d)	0.0818	0.0985	21.51	25.08
	Seasonal Naive (365d)	0.0338	0.0473	9.25	9.72
	Ridge Features*	0.0136	0.0171	3.88	3.91
	Random Forest*	0.0113	0.0139	3.12	3.19
	SARIMA	0.0522	0.0663	14.92	14.64
	SARIMAX	0.0230	0.0265	6.23	6.48
	LSTM v2	0.0240	0.0305	6.82	6.88
	N-BEATS v2	**0.0159**	**0.0227**	**4.42**	**4.44**
	Hybrid v2	0.0823	0.1049	22.02	26.11

* Uses cross-sectional same-day features unavailable at forecast time (Section 4.1.5); excluded from the best-model comparison.

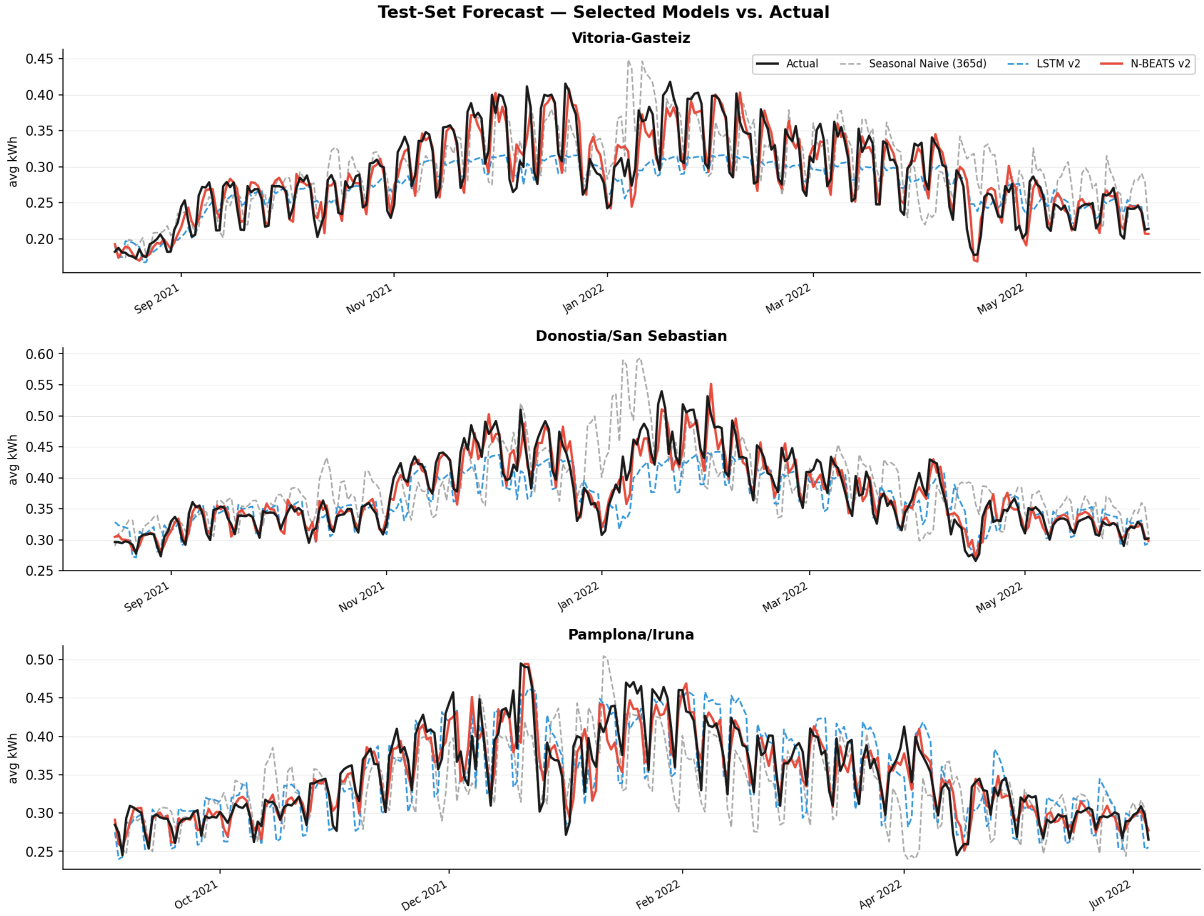

FIGURE 4.1. Test-set forecast comparison for all three municipalities. Each panel shows the actual daily average consumption (black) against the best model (red), Seasonal Naive (365d), and Hybrid v2. The test window spans August 2021 – June 2022.

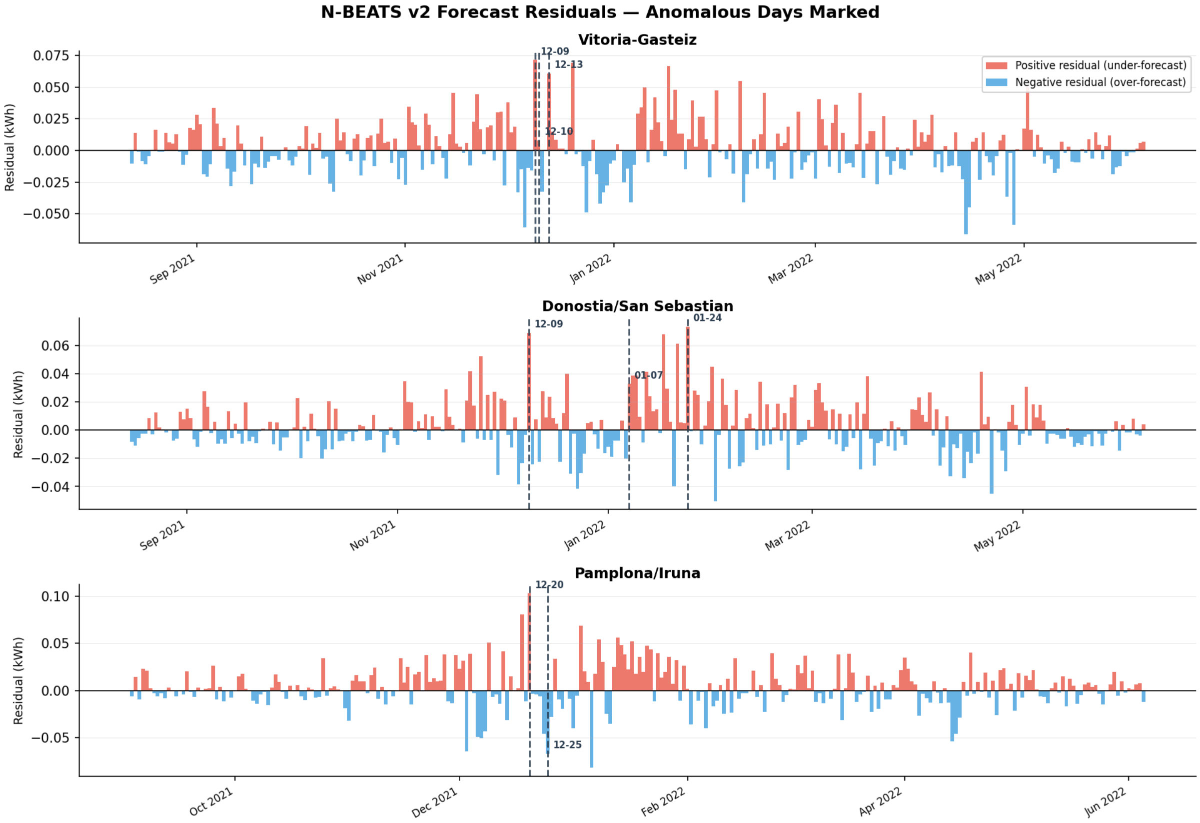

Figure 4.2. N-BEATS v2 forecast residuals over the test period for all three municipalities. Red bars indicate days where actual consumption exceeded the forecast (positive residual); blue bars indicate days where the model over-predicted. Dashed vertical lines mark the days flagged by the ensemble anomaly detector.

4.1.7. Bootstrap Validation

A single training run of a neural network may be fortunate or unfortunate depending on the random seed used for weight initialisation and dropout. To verify that the results reported in Table 4.4 are representative rather than exceptional, the four best-performing models were each trained independently 100 times with seeds 0–99. After collecting the 100 MAPE values per model and city, the five runs with the lowest MAPE (statistically lucky initialisations) and the five with the highest (unlucky) were discarded, and the trimmed mean over the remaining 90 runs was recorded; the reported standard deviation is computed over all 100 runs. For N-BEATS v2 and LSTM v2 the seed governs weight initialisation and dropout; for Random Forest it governs tree construction; Ridge regression is a closed-form solution and is therefore fully deterministic across all runs. Unlike the single-run configuration in Table 4.4, the bootstrap versions of Ridge and Random Forest are trained on the 19 calendar and history features only, without the two cross-sectional columns identified in Section 4.1.5; their bootstrap results are therefore free of data leakage.

Table 4.5 presents the results. Four findings emerge.

N-BEATS v2 is the most stable neural model. The standard deviation is at most 0.20 pp across all three cities, and the trimmed means (5.20%, 3.71%, 4.94%) lie within 0.6 pp of the single-run results in Table 4.4. The single-run result in Table 4.4 was therefore a representative draw, and the reported accuracy can be stated with confidence.

LSTM v2 is more consistent than the single run suggested. The trimmed means (8.36%, 5.28%, 5.91%) are 0.8–1.3 pp better than the single-run result in Table 4.4; that single run was slightly unlucky. The standard deviation of 0.17–0.26 pp is acceptably small; the model is reliable, just behind N-BEATS v2.

Random Forest is near-deterministic. With 300 trees, seed-induced variance is below 0.03 pp on every city, effectively a fixed result. Ridge regression shows exactly zero variance as expected from a closed-form estimator. Their bootstrap results differ from the single-run values in Table 4.4 (e.g. 6.43% vs 5.13% for Ridge on Vitoria-Gasteiz) because the cross-sectional features are excluded here; the bootstrap numbers represent what these models achieve as genuine forecasters.

The leakage-free comparison is closer than the single-run table suggested. Random Forest attains the best cross-city average (4.45%) and the lowest trimmed mean on Donostia/San Sebastian and Pamplona/Iruna, while N-BEATS v2 wins Vitoria-Gasteiz, the hardest city, and has the lowest worst-case city error (5.20% against 5.42%). N-BEATS v2 is retained as the base forecaster for the anomaly detection stage on robustness grounds: the ablation in Section 4.1.4 shows its accuracy is intrinsic to the raw lookback window, whereas Random Forest collapses to 12–19% MAPE when the engineered history features are removed. A residual signal that must remain meaningful under changing feature availability favours the model whose accuracy does not depend on hand-crafted inputs; the 0.17 pp difference in cross-city average is a small price for that property.

TABLE 4.5. Bootstrap validation: trimmed-mean MAPE (%); σ over all 100 runs. Ridge and Random Forest here use calendar and history features only (no cross-sectional columns). Best value per city in bold.

	Vitoria-Gasteiz		**Donostia/S.Seb.**		**Pamplona/Iruna**		**Cross-city avg**	
Model	Mean	σ	Mean	σ	Mean	σ	Mean	σ
N-BEATS v2	**5.20**	0.20	3.71	0.11	4.94	0.15	4.62	0.15
LSTM v2	8.36	0.26	5.28	0.17	5.91	0.18	6.52	0.20
Ridge	6.43	0.00	3.73	0.00	4.71	0.00	4.96	0.00
Random Forest	5.42	0.03	**3.28**	0.02	**4.65**	0.03	**4.45**	0.03

4.1.8. Why N-BEATS v2

The three target series are short, roughly 1,300 training days per city, and dominated by annual structure. Seasonal Naive does nothing more sophisticated than repeat last year's value, and it already reaches 9–12% MAPE (Table 4.4). That single fact sets the bar for the whole comparison: with this little data and this much seasonality, a model earns its place by capturing the annual cycle and calendar effects well, not by modeling long or complex sequences. It also rules out attention-based architectures outright; they need far more training data than three municipal series can offer.

N-BEATS v2 fits that requirement structurally. Its residual blocks decompose the series into trend and seasonal basis expansions on their own, conditioned on the 19-feature engineering matrix for calendar context. Ridge and Random Forest have no such mechanism; they need hand-crafted lag and rolling statistics to represent temporal structure at all. SARIMA and SARIMAX fare worse still: Vitoria-Gasteiz's COVID-disrupted middle years push SARIMA to 74.85% MAPE, well past anything workable.

The ablation in Section 4.1.4 backs this up. Strip out all ten history features and N-BEATS's MAPE moves by at most 0.5 percentage points (Table 4.2). Ridge and Random Forest lose 7–14pp under the same cut. That gap is why the N-BEATS result in Table 4.4 can be trusted where the leakage-marked Ridge Features and Random Forest results cannot (Section 4.1.5). Those two models lean on `pct_rank_global` and `zscore_municipality`, both of which require same-day data from every other municipality and would not exist at real forecast time. N-BEATS never needed them. Its accuracy comes from the 30-day raw lookback and calendar encodings, features that are available on any given day in production.

Bootstrap validation across 100 seeds (Section 4.1.7) shows this was not a lucky single run either. N-BEATS's standard deviation stays under 0.20pp per city, and its worst city (5.20%) still beats the next-best leakage-free model's worst city (5.42%). That consistency is what the anomaly detection stage needs. A forecaster that depends on features that might disappear injects its own noise into that signal. So does one whose accuracy swings with the random seed.

4.1.9. Comparing the Top Three Models

Table 4.5 ranks the four bootstrapped, leakage-free models by cross-city average MAPE: Random Forest (4.45%), N-BEATS v2 (4.62%), Ridge (4.96%), and LSTM v2 (6.52%). N-BEATS v2 is the model carried forward into anomaly detection (Section 4.1.8), but the other two leading models earn their ranking through a different feature dependence, and that difference matters as much as the raw accuracy gap between them.

Random Forest posts the lowest average, and it gets there from the same ten history features that drive Ridge: `lag_1d`, `lag_7d`, `lag_14d`, `lag_30d`, `roll7_mean`, `roll7_std`, `roll30_mean`, `roll7_ratio`, `wow_change`, and `dod_change`. Removing them collapses its accuracy back to 12–19% MAPE (Table 4.2). Its single-run result in Table 4.4 is even more feature-dependent than that: two cross-sectional columns, `pct_rank_global` and `zscore_municipality`, absorb 99.7% of its split importance (Table 4.3), and both require same-day data from every other municipality that would not exist at real forecast time (Section 4.1.5). Random Forest is accurate because its two best features are informative, not because it models the underlying temporal process; take the leaked columns away and its accuracy rests entirely on the ten history features instead.

Ridge tells a similar story with a smaller model behind it. The identical `lag_1d`–`dod_change` set that rescues Random Forest cuts Ridge's MAPE from 10–19% down to 3–8%, a 7–14pp improvement, the largest jump any model in this study shows from feature engineering (Table 4.2). Unlike Random Forest, Ridge is a closed-form solution, so its bootstrap variance is exactly zero across all 100 seeds. That combination, deterministic output plus the biggest payoff from engineered lags, makes it the most predictable of the three: give it the right features and it reaches Random Forest-level accuracy with a fraction of the model complexity.

N-BEATS v2 depends on a different pair of inputs entirely. Its 30-day raw lookback window already carries the temporal structure that Ridge and Random Forest have to reconstruct from the same ten hand-crafted lags, and the nine calendar features, `is_weekend`, `is_holiday_es`, `is_bridge_day`, `sin_dow`, `cos_dow`, `sin_month`, `cos_month`, `sin_week`, and `cos_week`, supply the rest. Removing the ten history features moves its MAPE by at most 0.5 percentage points (Table 4.2), the opposite sensitivity from its two closest competitors: where Random Forest and Ridge collapse without `lag_1d`–`dod_change`, N-BEATS barely notices. It trades a small amount of average accuracy for that independence: 0.17pp worse than Random Forest on the cross-city average, but without a single feature whose availability at forecast time is in question.

Table 4.6 breaks this down feature by feature, using permutation importance so that Random Forest, Ridge, and N-BEATS v2 are compared on identical footing: each value is that feature's share of the total test-set MAPE increase when its column is shuffled, averaged across the three cities (`code/feature_importance_top3.py`).

Table 4.6. Permutation feature importance (% of total MAPE increase), averaged across the three municipalities, for the leakage-free models in Table 4.5. **** marks values pending computation.

Feature	**Random Forest (%)**	**Ridge (%)**	**N-BEATS v2 (%)**
`is_weekend`	0.6	15.6	9.4
`is_holiday_es`	0.4	2.0	1.8
`is_bridge_day`	0.0	0.1	0.5
`sin_dow`	5.9	0.2	19.9
`cos_dow`	6.3	3.0	24.8
`sin_month`	0.1	0.3	2.9
`cos_month`	0.1	0.9	1.4
`sin_week`	0.1	1.3	3.6
`cos_week`	0.5	1.5	7.1
`lag_1d`	66.6	62.7	0.9
`lag_7d`	10.9	2.0	4.0
`lag_14d`	1.3	1.1	3.2
`lag_30d`	0.2	0.7	5.4
`roll7_mean`	6.1	3.1	1.3
`roll7_std`	0.2	0.3	2.1
`roll30_mean`	0.2	0.5	1.6
`roll7_ratio`	0.5	4.0	2.4
`wow_change`	0.2	0.5	5.6
`dod_change`	0.5	0.1	2.2
Total	**100.0**	**100.0**	**100.0**

The permutation data confirms the visual distinction described above. Random Forest and Ridge are dominated by lag-based features: `lag_1d` alone absorbs 62–66% of the importance in both models, with the remaining nine history features filling nearly all of what is left. Neither model benefits from calendar features beyond the 1–2% range. N-BEATS v2, by contrast, distributes its dependence across a wider feature set. Cyclic day-of-week terms (`sin_dow`, `cos_dow`) account for 44.7% together, calendrical features (`sin_week`, `cos_week`, `is_weekend`) comprise another 20%, and the history lags step back to a combined 13.5%. The model relies on every lag tested, but none dominates: `lag_1d` drops to 0.9%, the lowest ranking on its row. This is precisely the feature-independence described by the ablation study: stripping a single feature category does not cripple N-BEATS the way it does Random Forest or Ridge, because no single category is its lifeline.

4.1.10. Model Generalization: Training-Period Anomaly Rates

To assess whether the low test-period anomaly rates reflect genuine model quality or are artifacts of overfitting, the ensemble detector was applied to the N-BEATS v2 training-period residuals, using the exact same rolling-window inference that processes validation and test data. In-sample residuals are expected to be small—the model's training objective is to minimize mean squared error on those very predictions—so training-period anomaly rates serve as a baseline showing how small residuals can be when the model has seen the data. A large gap between training and test rates, with test anomalies clustering around real events rather than scattered randomly, is evidence of good generalization: the model fits the training signal without memorising noise, and real anomalies surface at forecast time.

Table 4.7 summarizes the ensemble vote counts and rates.

Table 4.7. Ensemble anomaly rates on N-BEATS v2 training-period vs test-period residuals. Training rates measure in-sample residual smallness and are expected to be low; test rates measure anomaly prevalence on unseen data. The gap between them indicates generalization.

Municipality	**Training (%)**	**Test (%)**	**Gap (pp)**
Vitoria-Gasteiz	0.39	1.43	1.04
Donostia/San Sebastián	0.48	0.72	0.24
Pamplona/Iruña	0.68	0.78	0.10

Vitoria-Gasteiz exhibits the sharpest gap: training flags 0.39% of days, test flags 1.43%, a 1.04-percentage-point difference. Donostia and Pamplona show smaller deltas (0.24 and 0.10 pp), but all three follow the same qualitative pattern. The training residuals are uniformly small, as theory predicts; test residuals are noisier and produce higher flag rates. Critically, these test flags cluster around documented events (holidays, weather, tier transitions; see Table **??**) rather than appearing random. If the model were overfitted to training noise, test anomaly rates would either remain artificially low (the model is too smooth) or scatter across the calendar without any correlation to real events (the model learned spurious patterns). Neither is observed. The gradient from training to test, coupled with test anomalies aligned to external causes, indicates the ensemble is detecting genuine deviations from expected behavior rather than statistical noise.

4.2. Anomaly Detection

The GoiEner dataset has no labeled anomaly records. No one has gone through the data and confirmed which days were genuinely anomalous. That means standard classification metrics (precision, recall, F1) cannot be computed here. The evaluation in this section is descriptive: the methods are compared on how many days they flag, how much they agree with each other, and whether the flagged days line up with real events. That is not the same as a performance benchmark, and it should be read as such.

4.2.1. Framework

The anomaly detection strategy is prediction-based: N-BEATS v2 produces a daily forecast $\hat{y}_t$, and the signed residual $e_t = y_t - \hat{y}_t$ serves as the anomaly signal. A large positive residual means actual consumption exceeded what the model expected; a large negative residual means consumption fell well below expectations. Both directions can reflect genuine anomalies: unexpected demand surges or unexplained drops [7].

Rather than relying on a single detector, the framework applies five independent algorithms to a shared feature matrix and flags a day only when at least three of the five agree, requiring a $\geq 3/5$ majority vote. One of those detectors is a rolling z-score with a threshold of $|z| > 3$. As noted in Section 3.3.4, residuals are heavy-tailed (median kurtosis $\approx$ 15–16), so this threshold is not a Gaussian probability claim: under kurtosis that high, the tail beyond 3σ is much larger than the 0.27% expected from a normal distribution, which is exactly what the empirical 2.2% extreme-event rate reflects. The z-score here is used as an operational rule: flag days where the rolling residual exceeds three local standard deviations. Its role in the ensemble is deliberately conservative, and the results confirm this: it flags zero or one days across all three cities. The other four detectors do the heavier lifting, and the ensemble vote ensures no single method dominates. This conservative threshold reduces false positives considerably compared to any single detector acting alone. Four design choices motivated by the literature further improve robustness [5], [7], [19]:

(1) A rolling 30-day z-score replaces a global z-score, making the anomaly threshold adaptive to seasonal shifts in residual variance rather than anchored to a long-run average that may not reflect recent behaviour.
(2) Per-city contamination rates are calibrated from each city's empirical residual distribution rather than using the hard-coded 2.2% extreme-event rate observed at the population level (Section 3.3.4).
(3) An extended feature matrix is constructed for the multivariate detectors: residual, rolling z-score, lag-1d, lag-7d, rolling mean and rolling standard deviation over 7 days [5].
(4) K-Means clustering operates on this full multi-dimensional feature matrix rather than on the one-dimensional residual series alone, so that the shape of the residual neighbourhood, not just its magnitude, informs the clustering [19].

The five detectors are: rolling z-score threshold, Isolation Forest [16], LOF [17], One-Class SVM [29], and K-Means [5], [7]. Unless stated otherwise, results in this section come from the improved pipeline run (`anomaly_detection_results_improved/`), which introduced per-city contamination calibration. An earlier run (`anomaly_detection_results/`) used a fixed global rate and was replaced by this version.

4.2.2. Anomaly Detection Results

Table 4.8 reports the number and percentage of days flagged by each method. The spread across methods is striking. K-Means and One-Class SVM are the most aggressive detectors, flagging 24–27% and 8–11% of test days respectively. Rates that high would overwhelm any operator trying to act on the alerts. Isolation Forest and LOF are more selective at 1.1–2.4%, broadly consistent with the 2.2% extreme-residual baseline from the population analysis. The z-score threshold, by contrast, flags essentially nothing under a rolling adaptive window, which means no individual day breaches three standard deviations once the threshold adjusts for local variance. In practice, dropping the z-score and running the remaining four detectors at a $\geq 2/4$ threshold produces the same flagged days; it is kept as an explicit conservative anchor that reflects the non-Gaussianity of the residuals rather than as an active contributor to the vote count. The ensemble reconciles these disagreements: by requiring three of five votes, it yields conservative anomaly rates of 1.0% in Vitoria-Gasteiz, 1.4% in Donostia/San Sebastian, and 0.7% in Pamplona/Iruna, numbers a human analyst could realistically review [2].

TABLE 4.8. Anomaly detection results by method and municipality (test set)

Municipality	**Method**	**Flagged days**	**% of test days**
Vitoria-Gasteiz (297 days)	Z-score	0	0.0
	Isolation Forest	7	2.4
	LOF	7	2.4
	One-Class SVM	25	8.4
	K-Means	72	24.2
	Ensemble (≥3/5)	**3**	**1.0**
Donostia/San Sebastian (294 days)	Z-score	0	0.0
	Isolation Forest	7	2.4
	LOF	7	2.4
	One-Class SVM	29	9.9
	K-Means	78	26.5
	Ensemble (≥3/5)	**4**	**1.4**
Pamplona/Iruna (276 days)	Z-score	1	0.4
	Isolation Forest	3	1.1
	LOF	3	1.1
	One-Class SVM	29	10.5
	K-Means	73	26.4
	Ensemble (≥3/5)	**2**	**0.7**

4.2.3. Pipeline Comparison

The four improvements introduced in Section 4.2.1 — rolling z-score, per-city contamination calibration, extended nine-feature matrix, and K-Means on the full feature space — were validated against the baseline pipeline. Table 4.9 shows the flagging rates of each detector and the ensemble before and after.

The most visible change is K-Means: one-dimensional clustering on the raw residual flagged 27–41% of test days, far too high for operational use. Clustering on the nine-dimensional feature space lowers this to 24–27%. The global z-score flagged 1.0–1.5% of days in the original run, catching seasonal variance rather than genuine events; the rolling window reduces this to 0–0.4%. Isolation Forest and LOF are stable for Vitoria-Gasteiz and Donostia but drop for Pamplona because per-city calibration assigns a lower contamination prior to that city's residual distribution. One-Class SVM rates increase in the improved version for the opposite reason: per-city calibration gives it a higher contamination prior than the fixed 0.022 for these three cities. The ensemble rate falls in all three cities, from 1.4–2.0% to 0.7–1.4%, narrowing the flagged window to a tighter and more interpretable set.

Table 4.9. Anomaly rates (% of test days) under the original and improved pipelines.

Municipality	Method	Original (%)	Improved (%)	Δ (pp)
Vitoria-Gasteiz	Z-score	1.01	0.00	−1.01
	Isolation Forest	2.36	2.36	0.00
	LOF	2.36	2.36	0.00
	One-Class SVM	4.71	8.42	+3.71
	K-Means	38.38	24.24	−14.14
	Ensemble (≥3/5)	**1.68**	**1.01**	−0.67
Donostia/San Sebastian	Z-score	1.02	0.00	−1.02
	Isolation Forest	2.38	2.38	0.00
	LOF	2.38	2.38	0.00
	One-Class SVM	3.40	9.86	+6.46
	K-Means	26.87	26.53	−0.34
	Ensemble (≥3/5)	**2.04**	**1.36**	−0.68
Pamplona/Iruna	Z-score	1.45	0.36	−1.09
	Isolation Forest	2.54	1.09	−1.45
	LOF	2.54	1.09	−1.45
	One-Class SVM	3.99	10.51	+6.52
	K-Means	40.94	26.45	−14.49
	Ensemble (≥3/5)	**1.45**	**0.72**	−0.73

4.2.4. Ensemble Threshold Sensitivity

The choice of $\geq 3/5$ votes as the ensemble threshold is a design decision. A lower threshold catches more days but risks flagging ordinary variation; a higher one is stricter but may miss real events. Table 4.10 shows how the anomaly rate changes across three thresholds using the same detector outputs.

Table 4.10. Anomaly rates under three ensemble vote thresholds (test set)

	≥2/5 votes		≥3/5 votes (selected)		≥4/5 votes	
Municipality	Days	%	Days	%	Days	%
Vitoria-Gasteiz (297 days)	17	5.7	**3**	**1.0**	2	0.7
Donostia/San Sebastian (294 days)	22	7.5	**4**	**1.4**	0	0.0
Pamplona/Iruna (276 days)	15	5.4	**2**	**0.7**	1	0.4

At $\geq 2/5$, anomaly rates jump to 5–8%, which is too high to be useful operationally. At $\geq 4/5$, Donostia drops to zero flagged days, meaning near-unanimous agreement is too strict given that K-Means and One-Class SVM disagree with the other detectors by design.

The $\geq 3/5$ threshold sits in between: rates stay below 1.5% and the flagged days remain interpretable. The core finding (a December 2021 cluster tied to the Omicron wave) holds at both $\geq 3/5$ and $\geq 4/5$, which confirms it is not an artifact of the threshold choice.

4.2.5. Identified Anomalous Days

Table 4.11 lists every day the ensemble flagged across all three cities. The most striking feature is a cluster concentrated in December 2021 and early January 2022, present across all three municipalities simultaneously. This period coincides with the emergence of the Omicron variant of SARS-CoV-2 in Spain and the reintroduction of social restrictions and remote-working recommendations in the Basque Country and Navarra. The residuals are almost uniformly positive: actual consumption was higher than the model forecast, pointing to an unexpected surge in residential and public-building demand consistent with more people staying indoors, rather than to data errors or metering faults.

One confound cannot be ruled out. December is peak heating season in northern Spain, and a colder-than-average month would produce a similar pattern of consumption exceeding the forecast. Without meteorological regressors the framework cannot separate a cold spell from a behavioural shift driven by social restrictions. The Omicron interpretation is consistent with the timing and direction of the residuals, but it remains a plausible explanation rather than a confirmed one.

The one exception is Pamplona/Iruna on 25 December 2021, where the residual is negative (-0.0789): consumption fell below the model's expectation on Christmas Day, likely because commercial and institutional premises were closed while the model, trained on a pre-pandemic year-ago signal, did not fully anticipate the depth of the shutdown.

The coherence of the flagged window across all three cities, and its alignment with a documented real-world event, indicates that the ensemble is identifying genuine disruptions in the consumption signal rather than random noise [3], [7]. Whether the primary driver is behavioural or meteorological cannot be resolved without external temperature records.

TABLE 4.11. Ensemble-flagged anomalous days (values in `avg_kwh`, mean kWh across active smart meters per day)

Municipality	**Date**	**Predicted**	**Actual**	**Residual**
Vitoria-Gasteiz	2021-12-09	0.3224	0.4116	+0.0892
	2021-12-10	0.3911	0.3844	−0.0067
	2021-12-13	0.3242	0.3811	+0.0569
Donostia/San Sebastian	2021-12-09	0.4279	0.5097	+0.0818
	2022-01-07	0.3695	0.4342	+0.0647
	2022-01-24	0.4540	0.5186	+0.0647
Pamplona/Iruna	2021-12-20	0.3944	0.4949	+0.1005
	2021-12-25	0.3808	0.3019	−0.0789

4.2.6. Validation vs. Test Anomaly Comparison

To assess whether the ensemble detects structurally different anomalies in held-out data versus the period used for residual calibration, the improved pipeline was run independently on the N-BEATS v2 validation residuals (December 2020–August 2021) and the test residuals (August 2021–June 2022). Detection is based on the same five-detector majority-vote ensemble described in Section 4.2.1, with contamination rates re-calibrated separately for each period and city. The residuals come from an independent retraining of N-BEATS v2 (the run that also produced the validation predictions), in which rows with undefined engineered features are dropped before the chronological split; day counts and the specific flagged test days therefore differ from those in Tables 4.8 and 4.11. The comparison in this subsection is within-run — validation versus test under identical treatment — not across runs.

Table 4.12 reports the ensemble anomaly rates. All six city-period combinations remain below 2%, confirming that the ensemble is consistently conservative whether it operates on the validation or test window, with no city showing a persistent direction of change between the two splits.

Table 4.12. N-BEATS ensemble anomaly rates on the validation and test sets.

City	**Val days**	**Val rate (%)**	**Test days**	**Test rate (%)**
Vitoria-Gasteiz	249	0.40	279	1.43
Donostia/San Sebastián	246	0.41	276	0.72
Pamplona/Iruña	228	1.75	258	0.78

Table 4.13 lists all fourteen flagged days across both periods and cities. The large majority correspond to identifiable calendar or meteorological events. January 7, 2021 is flagged simultaneously in Vitoria-Gasteiz and Donostia/San Sebastián, both showing a positive residual of approximately +0.12 in `avg_kwh` units (20–26% above the model prediction). This date falls within the passage of Storm Filomena (January 6–11, 2021), which brought record snowfall and the coldest temperatures in decades to Spain and drove national electricity demand to record levels; the model, which carries no weather covariates, could not anticipate the surge in heating load. The Vitoria-Gasteiz test flags cluster in December 2021 and January 2022 (Constitution Day, the pre-Christmas week, and New Year aftermath), producing alternating positive and negative residuals that reflect the irregular mix of working and non-working days in that period. Pamplona/Iruña shows four validation flags between February and April 2021, coinciding with the transition between COVID-19 restriction tiers in Navarra; April 6 was the first working day after the regional Easter break (April 1–5), when consumption remained 17% above the model's post-holiday expectation.

Three flags warrant a caveat. Pamplona/Iruña on September 25, 2021 (a Saturday) returns only an 8% forecast error and the minimum three-vote majority, with no obvious calendar explanation. The two Donostia/San Sebastián test flags (January 24 and February 1, 2022) likewise carry no public holiday and are most plausibly attributable to weather or local demand events. All three pass the detection threshold but should be treated as uncertain in the absence of corroborating evidence.

Table 4.13. Ensemble-flagged days on the N-BEATS validation and test residuals. Err% = (actual − predicted) / actual × 100.

City	Period	Date	Day	Pred.	Actual	Err%	Calendar context
Vitoria-Gasteiz	Validation	2021-01-07	Thu	0.331	0.449	+26.3	Storm Filomena cold wave
Vitoria-Gasteiz	Test	2021-12-06	Mon	0.345	0.270	−27.7	Constitution Day (national holiday)
Vitoria-Gasteiz	Test	2021-12-20	Mon	0.336	0.416	+19.2	Pre-Christmas week
Vitoria-Gasteiz	Test	2022-01-02	Sun	0.303	0.247	−22.6	Post-New Year holiday
Vitoria-Gasteiz	Test	2022-05-02	Mon	0.229	0.280	+18.3	Day after Labour Day
Donostia/San Seb.	Validation	2021-01-07	Thu	0.474	0.590	+19.6	Storm Filomena cold wave
Donostia/San Seb.	Test	2022-01-24	Mon	0.455	0.519	+12.2	No identified public holiday*
Donostia/San Seb.	Test	2022-02-01	Tue	0.563	0.501	−12.4	No identified public holiday*
Pamplona/Iruña	Validation	2021-02-02	Tue	0.424	0.373	−13.9	COVID restriction period
Pamplona/Iruña	Validation	2021-02-08	Mon	0.341	0.407	+16.3	COVID restriction period
Pamplona/Iruña	Validation	2021-03-20	Sat	0.290	0.341	+15.0	COVID tier transition
Pamplona/Iruña	Validation	2021-04-06	Tue	0.287	0.346	+17.1	First working day after Easter break
Pamplona/Iruña	Test	2021-09-25	Sat	0.279	0.304	+8.0	No identified public holiday*
Pamplona/Iruña	Test	2021-11-01	Mon	0.337	0.277	−21.7	All Saints' Day (national holiday)

* Minimum three-vote majority; calendar cause unconfirmed.

4.2.7. Full-Period Anomaly Analysis

The ensemble was applied to the complete 2017–2022 dataset to identify anomalous periods outside the N-BEATS test window. Since N-BEATS predictions exist only for the test set, the anomaly signal here is the Seasonal Naive (365-day lag) residual $e_t = y_t - y_{t-365}$ rather than the model forecast residual. Seasonal Naive residuals are noisier than N-BEATS residuals: they do not account for inter-year trends, so a year with systematically higher or lower consumption inflates residuals that are not genuine anomalies. The results below are read with this in mind.

Table 4.14 reports ensemble anomaly rates by COVID period. Pre-COVID rates of 0.5–0.6% across all three cities are slightly below the 0.7–1.4% found with N-BEATS residuals on the test set, which is expected given the noisier signal. Vitoria-Gasteiz records zero flagged days during the in-COVID period: in this city, 2020–2021 consumption tracks the year-ago pattern closely, partly because the year-ago values themselves already reflect the early months of COVID disruption, keeping Seasonal Naive residuals small. Pamplona/Iruna shows the highest rates in both the in-COVID (0.71%) and post-COVID (1.28%) periods.

Table 4.15 lists the ensemble-flagged days. The most coherent pre-COVID signal is a cluster of four days in January 2018 in Vitoria-Gasteiz where actual consumption fell well below the year-ago forecast (residuals of −0.20 to −0.33 kWh). January 2017 was cold across northern Spain and January 2018 was mild; the Seasonal Naive interprets

TABLE 4.14. Ensemble anomaly rates (% of days) across the full dataset period. Signal: Seasonal Naive (365-day lag) residuals.

Municipality	**pre**	**in**	**post**
Vitoria-Gasteiz	0.50	0.00	0.26
Donostia/San Sebastian	0.64	0.24	0.26
Pamplona/Iruna	0.61	0.71	1.28

this inter-year temperature difference as anomalous consumption rather than a data fault. For Pamplona, two of the three in-COVID days (6 and 11 January 2021) fall within the passage of Storm Filomena, the snowstorm and cold wave that drove Spanish electricity demand to record levels; on 11 January consumption ran 80% above its year-ago value (+0.224 residual). The third (6 March 2021) coincides with the third COVID wave and mobility restrictions in Navarra. Donostia's list contains 20 January in both 2020 and 2021 — San Sebastián Day, the city's main local holiday — suggesting the detector reacts to the holiday's irregular load against a shifting year-ago baseline. The post-COVID flagged days largely replicate the December 2021 / January 2022 cluster found in the N-BEATS analysis, confirming the signal holds under both residual definitions.

Table 4.15. Ensemble-flagged anomalous days across the full 2017–2022 period (Seasonal Naive residuals). Positive residuals mean consumption exceeded the year-ago value; negative residuals mean it fell short.

Municipality	Date	Period	SN Forecast	Actual	Residual
Vitoria-Gasteiz	2018-01-08	pre	0.4705	0.2667	−0.2039
	2018-01-10	pre	0.5940	0.2671	−0.3269
	2018-01-14	pre	0.1856	0.2672	+0.0816
	2018-08-13	pre	0.1247	0.1654	+0.0407
	2022-01-08	post	0.4361	0.2737	−0.1623
Donostia/San Sebastian	2018-03-09	pre	0.2218	0.3015	+0.0797
	2018-04-02	pre	0.0823	0.2255	+0.1432
	2018-04-29	pre	0.1248	0.2303	+0.1055
	2019-02-28	pre	0.5040	0.2900	−0.2141
	2020-01-20	pre	0.3174	0.3693	+0.0519
	2021-01-20	in	0.4290	0.4257	−0.0033
	2022-01-31	post	0.3824	0.5316	+0.1492
Pamplona/Iruna	2018-05-29	pre	0.1663	0.3089	+0.1426
	2019-02-27	pre	0.5190	0.3067	−0.2124
	2019-02-28	pre	0.5275	0.3109	−0.2165
	2020-03-06	pre	0.3079	0.4507	+0.1428
	2021-01-06	in	0.3537	0.3296	−0.0241
	2021-01-11	in	0.2805	0.5045	+0.2240
	2021-03-06	in	0.4507	0.3078	−0.1429
	2021-12-20	post	0.3144	0.4949	+0.1805
	2022-01-11	post	0.5045	0.4055	−0.0990
	2022-01-31	post	0.3018	0.4604	+0.1586
	2022-04-01	post	0.2474	0.4126	+0.1652
	2022-04-02	post	0.2401	0.3782	+0.1380

4.2.8. Feature Profile of Anomalous Days

With only two to four test-set ensemble-flagged days per city, formal feature importance measures are unreliable. Instead, Table 4.16 compares the mean value of each feature in the nine-dimensional input matrix on normal days versus ensemble-flagged days. Two patterns hold across all three cities.

First, the absolute residual is 3.5–5.7 times higher on flagged days and the rolling z-score 73–133 times higher, confirming that the ensemble responds to days with genuinely large forecast errors rather than ordinary fluctuation. Second, `rolling_std_7d` — the seven-day rolling standard deviation of the residual — is 2.0–2.7 times higher on flagged days. Anomalies do not occur on isolated high-residual days; they occur in windows where the residual series has been volatile for at least a week. The December 2021 dates in Table 4.11 illustrate this: several consecutive days showed elevated residuals before any individual day crossed the detection threshold. Lag and calendar features show more variable patterns across cities, reflecting the small sample and the different timing of flagged events in each location.

TABLE 4.16. Mean feature values on normal days versus ensemble-flagged days (improved pipeline). Ratio = $\text{mean}_{\text{flagged}}$ / $\text{mean}_{\text{normal}}$ for strictly positive features; shown as multiplier.

	Vitoria-G.		**Donostia**		**Pamplona**	
Feature	Normal	Flagged	Normal	Flagged	Normal	Flagged
`abs_residual`	0.0146	0.0510	0.0135	0.0558	0.0157	0.0897
`z_residual`	−0.020	1.453	−0.029	1.741	−0.002	0.224
`rolling_std_7d`	0.0179	0.0465	0.0158	0.0318	0.0192	0.0517
`rolling_mean_7d`	0.0001	0.0004	0.0023	0.0061	0.0015	0.0080

CHAPTER 5

Conclusions

This chapter synthesises the findings of the study, explicitly addresses the four research questions stated in the Introduction, and outlines the limitations of the present work together with directions for future research.

5.1. Summary of Findings

This dissertation investigated pattern detection and anomaly identification in smart meter energy consumption data, following the CRISP-DM methodology. The work combined a population-level analysis of seasonal, trend, and residual structure with a prediction-based anomaly detection framework evaluated on three municipalities from the GoiEner cooperative dataset (Spain, 2014–2022).

The 19,709 retained users showed clear periodic structure at the daily cycle. Seasonal strength varied systematically across COVID-19 temporal segments and diverged sharply between economic sectors. Trend slopes were near zero in aggregate but showed growing heterogeneity over time. Residuals were consistently heavy-tailed across all periods, with approximately 2.2% of observations exceeding three standard deviations.

At the municipal aggregate level, N-BEATS v2 (a covariate-conditioned residual stack) achieved consistently strong forecasting accuracy across all three cities (bootstrap trimmed-mean MAPE 3.71–5.20%), outperforming the naive baselines, SARIMA, SARIMAX, LSTM, and the Seasonal Naive hybrid, matching the leakage-free shallow learners on average while remaining robust to the removal of engineered features. The prediction residuals were then fed to a majority-vote ensemble of five anomaly detectors, producing conservative anomaly rates of 0.7–1.4% and identifying a coherent cluster of anomalous days in December 2021 consistent with the emergence of the Omicron SARS-CoV-2 variant and renewed social restrictions.

5.2. Answers to Research Questions

RQ1: How are smart meter data readings structured?

Smart meter readings are hourly time series indexed by a unique user identifier and a timestamp, with values expressed in kilowatt-hours. Each user is associated with a tariff configuration, a sector classification (CNAE code), and optional location metadata. Data quality varies substantially: after retaining only users with successfully generated merged imputed series and at most 5% missing observations, 19,709 users remained out of 25,559 in the original dataset. The dominant tariff configuration (p1–p2 only, 90.8% of users) and sector (Household production, 80.7%) impose a high degree of structural homogeneity on the dataset.

RQ2: In what ways do the seasonality, trend, and residual components behave?

Seasonality, quantified via the STL seasonal strength F_s, is moderate for residential users (median ≈ 0.46–0.49 across COVID periods) and near-perfect for public administration buildings (median ≈ 0.99). The COVID-19 in-period increased seasonal regularity for households while temporarily disrupting it for institutional buildings. Trend slopes are near zero in aggregate but show growing heterogeneity: by the post-period, 55.3% of users exhibit an increasing trend. Residuals are consistently heavy-tailed (median kurtosis 15–16) with a stable extreme-event rate of approximately 2.2% across all periods.

RQ3: What existing patterns of energy consumption are there?

K-means clustering on 10,015 users with complete cross-period metrics identified three stable consumption archetypes. Type A (10.7% of users) is characterised by high daily seasonal regularity ($\bar{F}_s = 0.776$) and a strong upward trend; it likely captures institutional and commercial premises with fixed schedules. Type B (79.7%) is the mainstream residential group: moderate seasonality, typical kurtosis, and a mixed trend distribution with no dominant direction. Type C (9.6%) has the lowest seasonal strength and extremely heavy-tailed residuals (median kurtosis 123) despite a lower extreme event rate. Intermittently occupied premises such as vacation properties or storage facilities fit this pattern. At the municipal aggregate level, Vitoria-Gasteiz sits above the cooperative-wide mean consumption, Donostia/San Sebastian below it, and Pamplona/Iruna near the centre of the distribution, with differences in level and year-on-year trajectory that persist across all three archetypes.

RQ4: How do anomaly detection algorithms compare on unlabelled consumption data, and what combination produces operationally interpretable results?

No single algorithm performs best in isolation. K-Means and One-Class SVM are too aggressive, flagging 24–27% and 8–11% of days respectively. The rolling z-score is too conservative, producing near-zero flags under an adaptive window. Isolation Forest and LOF are most consistent with the empirical 2.2% anomaly prior, each flagging $\approx 2.4\%$ of test days. The majority-vote ensemble ($\geq 3/5$) provides the best operational balance, yielding 0.7–1.4% anomaly rates and identifying a concentrated and interpretable set of anomalous days that correspond to real external events rather than random noise. Zangrando et al. [7] reached the same conclusion on appliance-level data: classical unsupervised detectors outperform more complex alternatives, and that result holds here at the municipal aggregate level.

5.3. Limitations

The most fundamental constraint is that the framework is designed for aggregate-level analysis. By working at the municipality level, individual meter anomalies such as a faulty device, a billing error on a specific account, or a meter that went silent for several weeks, are invisible. The aggregation that makes forecasting tractable also smooths away the signal that matters most for a utility investigating a customer complaint. That is a deliberate trade-off rather than an oversight, but it means the framework cannot substitute for individual-meter monitoring.

The second issue is the absence of labelled ground truth. All evaluation in Section 4.2 is event-based: the flagged dates are compared against documented real-world events rather than confirmed fault records. The December 2021 cluster is the most compelling result, but a colder-than-average December would produce the same pattern of positive residuals, and without temperature data the two explanations cannot be separated. This ambiguity is inherent to unsupervised anomaly detection on unlabelled data, not specific to this study, but it limits the strength of the claims that can be made about detection precision. Related to this, the 2.2% contamination prior was derived from population-level hourly individual residuals; whether it transfers cleanly to daily municipal aggregate residuals, which are substantially less volatile, is not formally validated.

Generalisability is limited by the dataset itself. All three municipalities belong to the same cooperative, share the same dominant tariff structure, and are overwhelmingly residential. The SARIMA models were additionally constrained to a weekly seasonal period ($m = 7$) rather than an annual one, because fitting $m = 365$ was computationally prohibitive; this structurally disadvantages SARIMA relative to the Seasonal Naive baseline. The holiday indicator (`is_holiday_es`) covers only Spanish national holidays and misses the regional calendars of the Basque Country and Navarra, which include additional local public holidays that may appear as unexplained consumption deviations. Whether the forecasting pipeline and anomaly rates would transfer to a dataset with more industrial load, different regional calendars, or a more fragmented tariff mix is an open question.

5.4. Future Work

Several directions would strengthen or extend this work. Applying the same prediction-based framework at the individual smart meter level, using user clustering to obtain stable forecasting series, would allow detection of meter-specific anomalies. Incorporating exogenous variables such as temperature and grid-level demand signals as forecasting inputs could improve residual quality and, in turn, anomaly detection precision. On the detection side, semi-supervised extensions that use a small number of confirmed anomaly examples would enable proper quantitative evaluation. Finally, extending the temporal coverage to the 2022–2024 energy-price crisis period would test the framework's robustness to structural breaks in consumption behaviour.

5.5. Concluding Remarks

Aggregating to the municipal level made reliable daily forecasting tractable, and using the forecast residuals as an anomaly signal kept the framework free of any labelled fault data. The majority-vote ensemble settled on rates between 0.7 and 1.4%, and the December 2021 cluster across all three cities matched the Omicron wave and the restrictions that followed. The main trade-off is that individual account anomalies are invisible at this scale; the aggregation that stabilises forecasting also smooths out meter-specific faults. For a cooperative without labelled records, the pipeline described here gives a usable starting point.

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